

Inflation Inequality Across the Euro Area¹

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Abstract

This article documents inflation inequality across household categories in the euro area using harmonized HICP prices and household budget surveys. We construct country-level inflation indicators by income, age, and residence area, aggregate them to the euro area, and assess recent fiscal interventions using counterfactual price indices. During the 2021–2023 inflation episode, the cumulative inflation gap between the first and fifth income quintiles is 0.6 point in the euro area. Over the same period, cumulative inflation for households aged 60 or over exceeds that for households under 30 by 1.3 points, while the cumulative rural–city gap is 1.0 point. Counterfactual indices indicate that household price policies, including tariff shields, VAT and charge cuts, fuel rebates, and price caps, lower average cumulative inflation by 1.9 points and reduce the income-quintile inflation gap by 1.1 points in the euro area.

Keywords: Inflation, Distribution of Income, Unconventional Fiscal Policy

JEL codes: E31, D30, C43

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Introduction

This paper examines the effect of the recent inflation surge on euro-area households. Inflation rates in the euro area rose sharply from mid-2021 onward, displaying considerable dispersion across countries and socio-demographic groups. Central banks target aggregate inflation, but the welfare cost of a given aggregate inflation rate is heterogeneous across households. High inflation disproportionately affects households whose consumption baskets are more exposed to rapidly rising prices, in particular households with higher expenditure shares on energy and food. The 2021–2023 episode was a large relative-price shock, initially driven by imported energy and commodity prices.

Inflation inequality requires both relative-price changes and heterogeneous expenditure shares. If all prices rose proportionally, fixed-basket inflation would be identical across households. Inequality emerges when relative-price movements interact with differences in household consumption baskets. We examine how recent energy-price shocks and the fiscal response shaped household inflation exposure across the euro area. Specifically, we construct monthly inflation indices by income, age, and residence area for each euro-area country and aggregate them using Eurostat’s HICP methodology. We also use household-level expenditure records to measure within-group dispersion and construct counterfactual price indices that remove the main price-based fiscal interventions of 2021–2023.

Our main results are the following. Inflation inequality varies substantially across euro-area countries. Over 2021–2023, the cumulative Quintile 1–Quintile 5 gap reaches several points in the Baltic countries and Ireland, but is small in France and Spain. In the euro area, the cumulative income gap is 0.6 point; the corresponding age and rural–city gaps are 1.3 and 1.0 points. The inflation burden is 3.1, 10.2, and 7.4 percent of net income for the first quintile in 2021, 2022, and 2023, compared with 2.0, 5.7, and 4.1 percent for the fifth quintile. Older and rural households experience higher inflation during the rise in energy prices. In household-level regressions, age and residence are the strongest correlates during the energy shock. The rural–urban breakdown is large in 2021–2022 but disappears in 2023, whereas the age gradient remains visible in 2022–2023; the association with income changes sign over time. Finally, household price policies lower average cumulative inflation by 1.9 points and reduce the aggregate income-quintile gap by 1.1 points in the direct index comparison. A complementary input–output simulation shows how energy-sector disturbances reach households independently of the policy counterfactual: direct exposure accounts for 0.7 point of the simulated 1.0-point income gap, and indirect propagation through production networks accounts for the remaining 0.3 point. Together, these decomposi-

tions show that energy shocks generate inequality both at the point of household purchase and upstream through firms' input costs, whereas the fiscal response primarily cushioned the direct household-energy channel.

The literature measures household inflation by combining price changes with differences in expenditure patterns. Early contributions reweight aggregate price indices for demographic groups (Hobijn and Lagakos, 2005), while studies using household-level expenditure or scanner data document substantial dispersion within those groups (Argente and Lee, 2021, Kaplan and Schulhofer-Wohl, 2017). More recently, Jaravel (2024) constructs long-run distributional consumer price indices for the United States, closely following the BLS methodology. For Europe, Pallotti et al. (2024) combines consumption, income, and balance-sheet data to measure welfare losses across euro-area households. Building on Charalampakis et al. (2022), we construct monthly indices by income, age, and residence area for all 20 euro-area countries. Unlike existing cross-country indicators, our group-specific baskets aggregate to official HICP item weights. Because we assign a common price index to all households within each COICOP category, our measures capture expenditure-share heterogeneity but not differences in prices paid, product quality, or within-category substitution (Sangani, 2026).

A related literature evaluates the fiscal response to the recent inflation surge. Using household microsimulation models, Pallotti et al. (2024) and Amores et al. (2025) show that fiscal interventions reduced welfare losses and inequality, although broad price measures were less targeted than income support. Using detailed UK household data, Levell et al. (2026) highlight a trade-off between targeting and efficiency: energy price subsidies better protect households with high exposure to energy-price shocks, including within income groups, but distort consumption choices, while transfers preserve price signals but may be less well targeted.

Bonnet et al. (2025) show, for France, that income-targeted transfers are more effective than fuel price subsidies at limiting welfare losses relative to household income. We instead reconstruct monthly no-policy price paths at the COICOP-component level for all 20 euro-area countries. This approach identifies which products drove the distributional effect of price-based policies: lower electricity and gas prices reduce the Quintile 1–Quintile 5 inflation gap, while fuel measures partly offset that reduction because higher-income households spend more on private transport.

This paper makes two contributions. First, we construct harmonized household-group price indices that aggregate to official HICP weights for all 20 euro-area countries. These indices measure HBS budget shares at both the group and household levels while holding item-level price changes common across households. Second, we show that price-based fiscal measures reduce

inflation inequality in the euro area mainly by cushioning household-energy prices.

The remainder of the paper is organized as follows. Section 1 describes the data and methodology. Section 2 examines developments in inflation inequality in the euro area. Section 3 presents simulations of the heterogeneous effects of unconventional fiscal policies. Section 4 turns to household-level inflation heterogeneity.

1 Data and Methodology

This section explains how the household-group inflation indicators are built from the same accounting logic as the Harmonised Index of Consumer Prices (HICP). We first describe the data sources and harmonization challenges that motivate the calibration of household-group weights to official HICP weights.

1.1 Constructing Household-Group Price Indices

Headline HICP inflation represents the price change faced by an average household basket. We compute the corresponding price index for household categories: income groups, age groups, and residence areas. Prices are common within each product category, as in the official HICP, but the expenditure weights differ across household groups.

This analysis is based on publicly available data, primarily the HICP constructed by national statistical institutes in collaboration with Eurostat. The HICP offers a standardized measure of inflation, ensuring comparability across Member States. Eurostat aggregates national HICPs. Each country’s HICP is based on price changes for a fixed basket of goods and services, weighted according to household expenditure from country-specific sources, including national accounts, household budget surveys, and other national sources. The national indices are then aggregated into the euro-area HICP using country-specific weights that reflect their share of total household consumption expenditure.¹

To ensure the HICP reflects recent consumption patterns, Eurostat updates its weighting structure annually. The updated weights are implemented with the January release of the HICP and remain fixed throughout the calendar year, which is the basis for the within-year Laspeyres aggregation. Let $\mathcal{J}_y$ denote the set of product categories with official HICP weights in year y . Let $p_{j,y,m}$

¹The calculation ensures comparability by adhering to Regulation (EU) 2016/792 and consistent data collection practices. The HICP is compiled according to the European classification of individual consumption by purpose (ECOICOP v2)

denote the price index of product category j in month m of year y , and let $p_{j,y,0}$ denote the price reference for the HICP weight year, corresponding to the previous December. The official within-year Laspeyres index is obtained by aggregating unchained product indices with the annual HICP product weights:

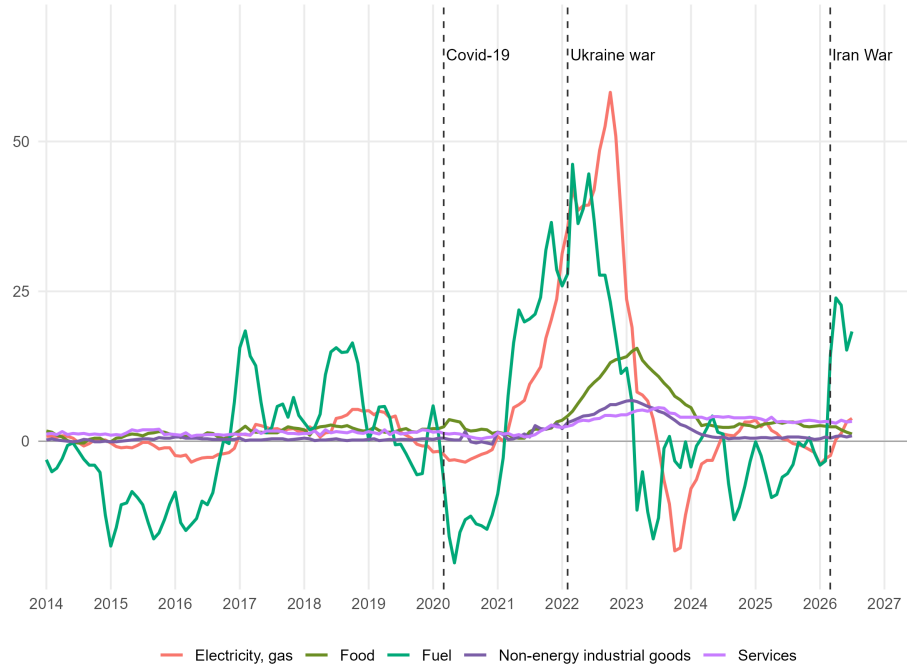
$$P_{y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,y}^{\text{HICP}} \cdot \frac{p_{j,y,m}}{p_{j,y,0}}. \quad (1)$$

The chained HICP index is then obtained by linking this within-year index to the previous December:

$$I_{y,m} = I_{y-1,12} \cdot P_{y,m}. \quad (2)$$

Figure 1 shows inflation developments across the main HICP components in the euro area since 2014. The series show marked volatility in energy and fuel prices. The sharp rise starting around mid-2021 and peaking in 2022 reflects the post-COVID recovery, supply-chain disruptions, and geopolitical tensions. After the 2022 peaks, inflation declined across most components. Energy prices dropped sharply, while food prices and other components stabilized more gradually.

Figure 1: Inflation developments in the euro area (year on year, in %)



Sources: Eurostat HICP euro-area aggregate; authors' calculations.

Consumption Data and Harmonization Challenges

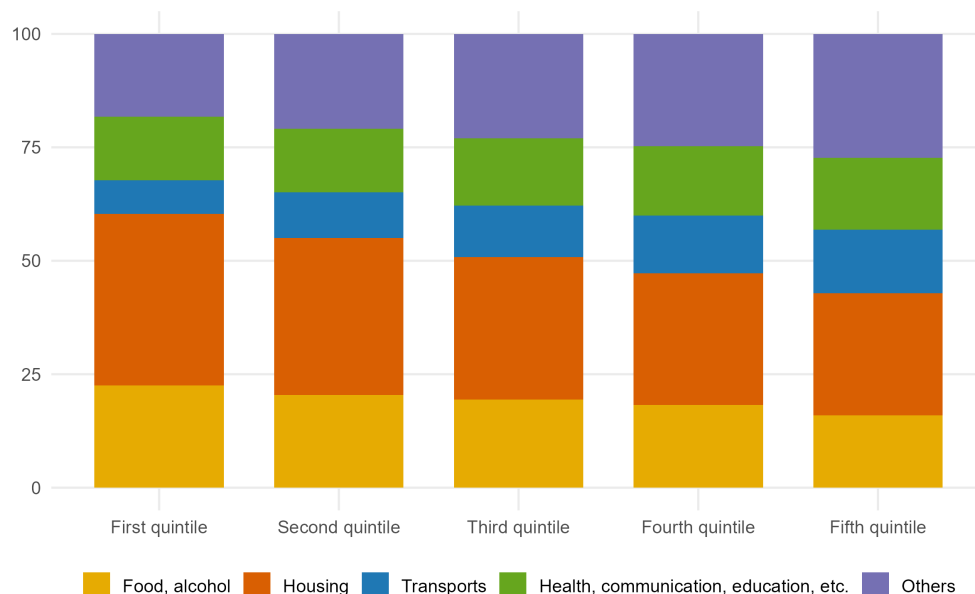
The heterogeneity of consumption patterns among households is well-documented, but the accurate measurement of household-level consumption continues to present significant challenges. Estimating HICP inflation by household category requires Household Budget Surveys (HBS), which are national surveys on consumer expenditure.

The Household Budget Surveys cover a large sample (approximately 150,000 households in the euro area). Households are required to document their expenditures on consumer goods and services over a designated period, typically two weeks. These surveys capture the values of expenditures across COICOP categories. HBS data are generally collected every five years.² The most recent complete HBS wave generally refers to 2020. Data for this wave were collected between 2018 and 2022 in most euro-area countries, with a few country-specific timing exceptions.

National statistical institutes in Italy and Spain also provide higher-frequency HBS data that can be used for country-specific extensions. Repeated HBS waves available for Spain suggest that, while behavioral changes over the course of a few years can be significant, point differences in the weights of the main consumption categories between low- and high-income households remain relatively stable (see Appendix A). Changes in overall consumption patterns may therefore have a limited effect on measured inflation inequality when consumption shares move in the same direction across income groups.

²See [Accardo et al. \(2023\)](#), who estimate developments in consumption inequality between survey waves.

Figure 2: Consumption baskets by income quintiles in the euro area (in % of total expenditures)



Notes: Structure of consumption expenditure by income quintile. The euro-area aggregate combines the 2020 country-level HBS income-quintile expenditures for all 20 countries and is renormalized to 100 within each quintile. Italy uses baskets constructed from Istat HBS microdata; see Appendix C.

Sources: Eurostat HBS, Istat HBS microdata, and authors' calculations.

While HBS expenditure shares are observed only at low frequency, our weighting approach combines household-specific HBS expenditure structures with monthly HICP inflation rates. As a result, aggregate consumption patterns can evolve continuously over the business cycle, reflecting changes in the HICP.

Figure 2 presents consumption baskets by income quintile. Lower-income households allocate a larger share of expenditure to essentials such as food and energy, increasing their exposure to price shocks in these categories. Housing and food account for a substantially larger share of expenditure among lower-income households. Energy-related expenditure for housing and private transport is also much higher in rural areas than in cities, reflecting differences in heating needs and mobility constraints.

HICP and HBS are not directly harmonized. HICP item weights are updated every year and may rely on national accounts or other national sources, while HBS expenditure structures are observed at lower frequency. Depending on the level of data available, HICP and HBS are therefore merged at the three-digit COICOP in each country. In some cases, data published directly by national statistical institutes can be used instead of relying only on the Eurostat dataset. This

approach is particularly valuable when the nationally published data provide greater granularity, recency, or frequency.³

The HBS contains many household characteristics that can be used to segment inflation exposure. In this paper, we focus on three dimensions: income, the age of the household reference person, and the area of residence.⁴ These variables make it possible to compare inflation experiences not only across the income distribution, but also across demographic groups and territorial contexts. Although imputed rent is outside the official HICP coverage, we retain it when constructing group-specific expenditure profiles; Appendix D explains this treatment and its implications for aggregation. Appendix Table D.1 also reports the sensitivity of the results to alternative rent mappings. For the euro area, the 2021–2023 income-quintile inflation gap in this robustness exercise is 0.9 point under our baseline mapping of HBS actual and imputed rents (041+042) to HICP actual rents (041), compared with -0.5 point when only HBS actual rents are mapped to HICP 041 and 1.3 points when rents are excluded altogether. The treatment of rents therefore matters for the magnitude, and sometimes the sign, of measured inflation inequality, which supports retaining imputed rents in the group expenditure profiles rather than redistributing their expenditure share mechanically across non-housing items. This analysis uses all available waves of the HBS since 1999 and matches them with the closest corresponding HICP year. Country-specific extensions based on national HBS microdata are described in the appendices when harmonized Eurostat inputs are incomplete for the required household breakdowns.

These data constraints motivate the weighting methodology below. A simple use of HBS expenditure shares would generally not aggregate back to the official HICP, because the HBS all-household basket does not necessarily coincide with the annually updated HICP item weights. We therefore start from the official HICP product weights and use HBS only to distribute each product’s expenditure across household groups. The Row and Column Adjustment Scaling (RAS) calibration preserves the relative HBS expenditure profiles while imposing the row and column margins required by the HICP construction.

³The `inflationinequality` R package allows the use of national data at 4-digit COICOP.

⁴In recent HBS waves, Eurostat generally requests the Degree of Urbanisation (DEGURBA) variable, built from a common European methodology based on a 1 km² population grid. It classifies residence areas into cities (densely populated areas), towns and suburbs (intermediate density areas), and rural areas (thinly populated areas), using comparable thresholds across EU countries: urban centers of at least 50,000 inhabitants and 1,500 inhabitants per km², urban clusters of at least 5,000 inhabitants and 300 inhabitants per km², and all remaining areas.

Group-Specific Weights

Let $\omega_{j,y}^{\text{HICP}}$ denote the official annual HICP weight of product category j in weight year y . Let g denote a household group, such as an income quintile, an age group, or a residence-area group. Let y' denote the HBS wave matched to HICP weight year y . Let $E_{g,y'}^{\text{HBS}}$ and $E_{\text{all},y'}^{\text{HBS}}$ denote total HBS consumption expenditure by group g and by all households, respectively.

The weights $\omega_{j,g,y}$ are obtained from the RAS calibration. The algorithm preserves the relative HBS expenditure profiles while making the expenditure-weighted average of group baskets reproduce the official HICP item weights. Appendix D describes the RAS algorithm and compares it with the relative-expenditure normalization.

The calibrated aggregate expenditure matrix $M_{j,g,y}$ is required to satisfy the normalized HICP item-share margins. Let $s_{g,y}$ denote the share of total household consumption expenditure accounted for by group g in weight year y :

$$s_{g,y} = \frac{E_{g,y'}^{\text{HBS}}}{E_{\text{all},y'}^{\text{HBS}}}, \quad \sum_g s_{g,y} = 1. \quad (3)$$

Define the HICP weight normalized over the matched product set as

$$\bar{\omega}_{j,y}^{\text{HICP}} = \omega_{j,y}^{\text{HICP}} / \sum_{k \in \mathcal{J}_y} \omega_{k,y}^{\text{HICP}}.$$

The RAS calibration imposes:

$$\sum_{j \in \mathcal{J}_y} M_{j,g,y} = s_{g,y} \quad \text{for all } g, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}} \quad \text{for all } j. \quad (4)$$

The product margin, $\sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}$, ensures that the expenditure-weighted average of the group-specific weight assigned to each product reproduces the official HICP weight for that product. The group margin, $\sum_{j \in \mathcal{J}_y} M_{j,g,y}$, fixes the total expenditure mass of each household group, so that groups can retain different product structures while aggregating back to the published HICP basket. The final group-specific HICP weight is obtained by normalizing the calibrated matrix within each group:

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}} = \frac{M_{j,g,y}}{\sum_{k \in \mathcal{J}_y} M_{k,g,y}}. \quad (5)$$

After aggregation, this calibration closely reproduces the official inflation rate (see Appendix D), and validation checks are reported in Appendix E.

The year of the HBS data y' and the HICP weight year y may therefore differ. To align the two sources, each HICP weight year y is matched with the most recent HBS wave available at or before that year. If no earlier HBS wave is available, the earliest HBS wave is used.

As a rule, HICP data are available on Eurostat at a level of detail corresponding to the 5-digit COICOP classification. However, certain product categories lack price data for specific periods. For instance, early-year price data for medical consultation services ("062") is often unavailable. To address these gaps, missing price indices are estimated based on the price developments of their parent product categories. In the case of "062", the price movements of the broader "health services" category ("06") are used to estimate the missing data.

Group-Level Consumer Price Indices

Let $p_{j,y,m}$ denote the Laspeyres price index of product category j in month m of year y . Let $p_{j,y,0}$ denote the price reference used for the HICP weight year y ; in the chained annual calculation this corresponds to the December price index of the previous year. The aggregation procedure is as follows: item indices are first unchained within the weight year, aggregated with annual expenditure weights, and then chained again.⁵ The group-specific within-year price index therefore combines unchained product indices with the group-specific weights defined above:

$$P_{g,y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,g,y} \cdot \frac{p_{j,y,m}}{p_{j,y,0}}. \quad (6)$$

The chained price index for household group g , denoted $I_{g,y,m}$, is then obtained by linking each within-year index to the previous December:

$$I_{g,y,m} = I_{g,y-1,12} \cdot P_{g,y,m}. \quad (7)$$

This notation keeps the two objects distinct: $\omega_{j,g,y}$ is the annual group-specific HICP weight of product j , while $I_{g,y,m}$ is the resulting monthly chained price index for group g .

2 Main Results

This section presents our main results. We focus on inflation inequality in the wake of the 2021–2023 energy crisis and also document long-run inflation inequality in the euro area.

⁵See Eurostat's `hicp` R package, which implements the Eurostat HICP aggregation.

2.1 Inflation Inequality in the Euro Area

2.1.1 Short-Run Inflation Inequality

Figure 3 shows that the post-COVID inflation surge translated into moderate income-based inflation inequality in the euro area. Inflation rates for the first and fifth income quintiles remain close throughout the 2021–2023 period. Energy and food shocks raised headline inflation sharply, but their effects on the relative inflation exposure of low- and high-income households largely offset once national consumption structures and country-level price dynamics are combined across all 20 countries.

This pattern is broadly consistent with recent evidence from [Bobasu et al. \(2023\)](#) and [Claeys et al. \(2022\)](#), although small differences remain because of our weighting scheme or treatment of imputed rents.

Figure 3: Short-run inflation inequality in the euro area, by income quintile (year on year, in %)



Notes: This figure reports year-on-year inflation rates for the first and fifth income quintiles in the euro area from January 2020 to July 2026. The country series are aggregated using annual HICP country weights.

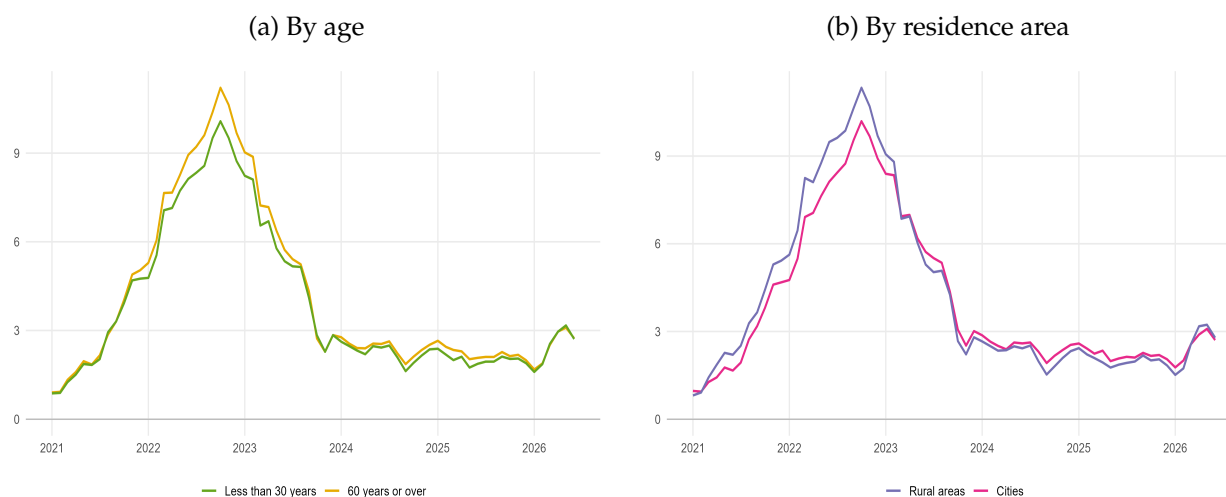
Sources: Eurostat HICP, HBS, authors' calculations

2.1.2 Inflation Inequality by Age and Residence Area

Figure 4 presents inflation rates by household category in the euro area, segmented by age and residence area. Cumulative inflation is 15.2 percent for households aged 60 or over and 13.9 percent for households under 30, yielding an age gap of 1.3 points. It is 15.3 percent in rural areas and 14.3 percent in cities, yielding a rural–city gap of 1.0 point. The monthly annual inflation rates in the figure show that both gradients vary over time and temporarily narrow or reverse as energy inflation recedes.

The residence-area differentials reflect differences in the structure of energy demand. Rural households tend to spend a larger share of their budget on transport fuels and home energy, partly because commuting distances are longer and the set of available transport substitutes is more limited. This raises their exposure when energy prices accelerate, but also makes their measured inflation fall faster when those price pressures unwind.

Figure 4: Inflation by household categories in the euro area (year on year, in %)



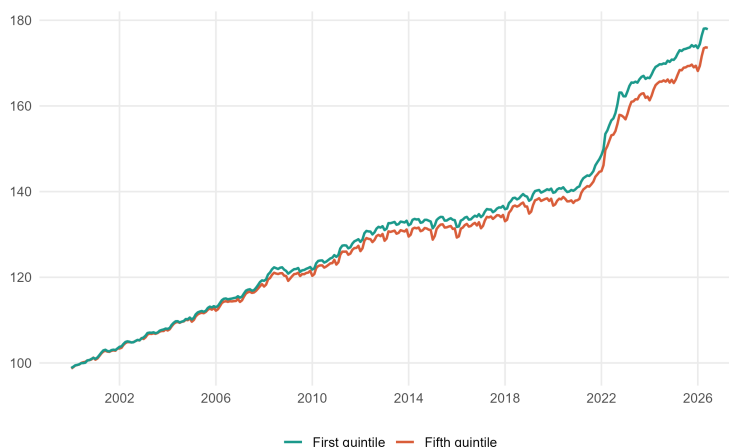
Notes: This figure reports year-on-year inflation rates in the euro area by age (panel a) and residence area (panel b), January 2020 to July 2026. The age and residence-area cumulative gaps cover 2021–2023, use June endpoints within those years, and are calculated as $100(P_{g,2023-06}/P_{g,2021-06} - 1)$ for each group before taking the difference. For age, cumulative inflation is 15.17 percent for households aged 60 or over and 13.90 percent for households under 30; the 60-or-over minus under-30 gap is therefore 1.27 points, rounded to 1.3. For residence, cumulative inflation is 15.27 percent in rural areas and 14.31 percent in cities; the rural–city gap is therefore 0.96 point, rounded to 1.0. All series use chained indices based in 2020 and annual HICP country weights.

Sources: Eurostat HICP, HBS, authors' calculations

2.1.3 Long-Run Inflation Inequality

Figure 5 shows a modest but persistent divergence between the first- and fifth-quintile price indices after 2008. The post-COVID episode produces the largest visible widening in the series since 2000.

Figure 5: Long-run inflation inequality in the euro area, by income quintile (index, 2000 = 100)



Notes: This figure shows monthly price indices for the first and fifth income quintiles in the euro area, January 2000 to July 2026. Indices are rebased to 2000 = 100.

Sources: Eurostat HICP, HBS, authors' calculations

This pattern contrasts with the larger long-run gaps documented for the United States in [Jaravel \(2024\)](#). Several methodological differences may account for the contrasting results. First, the euro-area indices reported here aggregate national indices and therefore average across countries whose price dynamics and fiscal responses can offset one another. Second, the euro-area calculation uses harmonized 3-digit COICOP categories, whereas the U.S. evidence exploits much finer product-level data. Aggregation within broad categories can conceal persistent differences in the prices paid by different income groups, which are captured in the U.S. data.

2.2 Heterogeneous Effects of the 2022–2023 Energy Shock Across Countries

The 2021–2023 inflation episode was marked by a large energy-price shock, especially in natural gas markets, following Russia's invasion of Ukraine. Although this shock was common to the euro area, its distributional effects differed across countries because energy mixes, import dependence, household consumption patterns, and fiscal responses were not uniform.

Table 1 documents this cross-country heterogeneity by reporting the cumulative 2021–2023

inflation gap between the first and fifth income quintiles. The Baltic countries and Ireland display large positive Quintile 1–Quintile 5 gaps, while several large economies show more compressed gaps.

The interaction between country-specific price dynamics, exposures, and fiscal policies and group-specific expenditure shares explains these differences. A price increase in a product category generates inflation inequality only to the extent that the affected category has different expenditure weights across household groups. Hence, similar income-quintile consumption profiles can still produce different inflation gaps when national energy prices, food prices, or fiscal cushioning measures evolve differently. Conversely, large aggregate shocks may generate limited inequality when price increases fall on categories with similar weights across quintiles or when policy measures dampen prices in essential energy components.

Table 1: Cumulative inflation gap between the first and fifth income quintiles, January 2021–December 2023

Country	Weight (2022, %)	Cumulative inflation	Inflation gap			
			2021–2023	2021	2022	2023
Germany	28.1	18.5	0.5	-0.4	0.9	0.4
France	20.4	14.6	0.7	0.3	0.1	0.1
Italy	16.3	18.6	1.3	0.1	1.5	0.4
Spain	11.0	16.5	-0.2	0.8	0.9	-0.7
Netherlands	5.4	20.1	0.6	-0.3	1.8	-0.9
Belgium	4.0	19.1	-0.3	1.2	1.9	-2.3
Austria	3.3	22.0	-0.4	-0.5	-0.2	0.5
Portugal	2.2	15.4	-0.4	0.7	0.1	-0.9
Greece	2.2	17.9	1.4	0.3	1.2	-0.5
Finland	1.9	13.2	-1.0	-0.2	-1.1	0.2
Ireland	1.5	17.8	4.2	0.4	2.8	1.1
Slovakia	0.8	28.4	2.1	-0.9	1.1	1.3
Croatia	0.7	24.7	-0.4	0.0	0.3	-0.3
Lithuania	0.6	34.3	3.6	0.2	3.3	0.4
Slovenia	0.4	21.2	1.3	0.1	0.5	1.4
Luxembourg	0.3	14.5	0.3	0.1	0.2	-0.1
Latvia	0.3	30.8	5.9	-0.2	4.6	2.4
Estonia	0.2	36.5	6.8	0.5	4.3	1.5
Cyprus	0.2	15.9	0.3	0.0	0.7	0.2
Malta	0.1	15.7	1.7	-0.4	0.2	1.2
Euro area	100.0	17.8	0.6	0.1	0.9	0.0

Notes. Cumulative changes compare January 2021 with December 2023; annual rates compare annual-average indices. The gap is Q1 minus Q5. Weights are 2022 EA20 HICP shares. Entries are percentage points. Sources: Eurostat, authors' calculations.

Housing energy and food raise the cumulative Quintile 1–Quintile 5 gap by 2.9 points over 2021–2023, while fuel and transport excluding fuel reduce it by 1.0 point (Table 2).

Table 2: Annual product-group contributions to the euro-area Q1–Q5 inflation gap

Product group	Weight (2022, %)	Cumulative inflation	Inflation gap			
			2021–2023	2021	2022	2023
Food and non-alcoholic beverages	16.6	4.2	1.4	0.1	0.6	0.6
Alcoholic beverage, tobacco and narcotics	4.3	0.8	0.4	0.1	0.1	0.2
Clothing and footwear	5.2	0.8	-0.3	0.0	0.0	-0.1
Housing (rentals and repairs)	12.2	2.1	-0.4	0.0	-0.2	-0.1
Water, electricity, gas and other fuels	6.3	3.4	1.9	0.4	1.6	0.1
Fuel	5.9	1.6	-0.5	-0.2	-0.4	0.0
Transport (excluding fuel)	8.9	1.9	-0.5	-0.1	-0.3	-0.2
Other	40.5	3.1	-1.4	-0.1	-0.5	-0.5
Total	100.0	17.8	0.6	0.1	0.9	0.0

Notes. The gap is Q1 minus Q5. Annual columns compare annual-average indices; cumulative columns compare January 2021 with December 2023. Weights are 2022 EA20 expenditure shares. Contributions sum to totals before rounding; no residual adjustment is used. Sources: Eurostat, authors' calculations.

2.3 Direct and Indirect Exposure to Energy Price Shocks

This section examines whether this exposure arose primarily from households' direct purchases of energy or from cost increases transmitted through production networks. We combine an input–output model with group-specific expenditure weights to decompose the simulated inflation effects into direct and indirect components.

Let $\mathbf{A}$ denote the matrix of intermediate-input coefficients in the euro area and let $\mathbf{s}$ be the sectoral price shocks. The Leontief price response is

$$\Delta \mathbf{p} = \underbrace{\mathbf{s}}_{\text{direct}} + \underbrace{\left[(\mathbf{I} - \mathbf{A}^T)^{-1} - \mathbf{I} \right] \mathbf{s}}_{\text{indirect}}. \quad (8)$$

The first term is the price change in the sectors initially shocked. The second term captures domestic and cross-country propagation through intermediate inputs. Let $\mathbf{M}$ denote the NACE–COICOP bridge from Cai and Vandyck (2020), normalized within each consumption category, and let $\mathbf{w}_g$ denote the expenditure weights of household group g . The corresponding direct, indirect, and

total household-price effects are

$$\Delta\pi_g^{\text{direct}} = \mathbf{w}_g^T \mathbf{M} \mathbf{s}, \quad (9)$$

$$\Delta\pi_g^{\text{indirect}} = \mathbf{w}_g^T \mathbf{M} \left[(\mathbf{I} - \mathbf{A}^T)^{-1} - \mathbf{I} \right] \mathbf{s}, \quad (10)$$

$$\Delta\pi_g = \Delta\pi_g^{\text{direct}} + \Delta\pi_g^{\text{indirect}}. \quad (11)$$

Distributional exposure is the difference between group effects. For income quintiles, for example, $\Delta\pi_{Q1-Q5} = \Delta\pi_{Q1} - \Delta\pi_{Q5}$, with the same decomposition into direct and indirect components.

Panel (a) imposes a 30 percent producer-price shock on mining and quarrying, refined petroleum, electricity, and gas in every euro-area country. Panel (b) imposes a 10 percent shock on agriculture. Both simulations use the 2022 FIGARO input–output matrix and 2022 HICP, RAS expenditure, and EA20 country weights. The initial shock $\mathbf{s}$ defines the direct effect; the remaining response generated by the price system captures domestic and cross-border cost propagation through intermediate inputs. We map the production price changes to consumption products and aggregate them using official HICP country and product weights for the average basket and expenditure weights for Q1 and Q5.

Table 3: Effects of energy- and food-price shocks by product group, euro area, 2022

Product group	Effect on average inflation	Effect on inflation gap	Direct effect on inflation gap	Indirect effect on inflation gap
<i>(a) Effect of a 30% energy-sector shock by product group, euro area, 2022</i>				
Food and non-alcoholic beverages	0.5	0.0	0.0	0.0
Alcoholic beverage, tobacco and narcotics	0.1	0.0	0.0	0.0
Clothing and footwear	0.1	0.0	0.0	0.0
Housing (rentals and repairs)	0.2	0.0	0.0	0.0
Water, electricity, gas and other fuels	2.2	0.7	0.5	0.2
Fuel	1.0	-0.5	-0.4	-0.1
Transport (excluding fuel)	0.7	-0.1	0.0	-0.1
Other	0.7	-0.1	0.0	-0.1
Total	5.7	0.2	0.1	0.1
<i>(b) Effect of a 10% food-price shock by product group, euro area, 2022</i>				
Food and non-alcoholic beverages	0.7	0.0	0.0	0.0
Alcoholic beverage, tobacco and narcotics	0.1	0.0	0.0	0.0
Clothing and footwear	0.0	0.0	0.0	0.0
Housing (rentals and repairs)	0.0	0.0	0.0	0.0
Water, electricity, gas and other fuels	0.0	0.0	0.0	0.0
Fuel	0.0	0.0	0.0	0.0
Transport (excluding fuel)	0.0	0.0	0.0	0.0
Other	0.1	0.0	0.0	0.0
Total	1.0	0.0	0.0	0.0

Notes. Entries are percentage-point effects. Panel (a) applies a 30 percent energy shock; panel (b) a 10 percent food shock. The simulation uses 2022 FIGARO and 2022 HICP, RAS, and EA20 country weights. Direct and indirect effects sum to the Q1–Q5 gap before rounding. Sources: Eurostat, HBS, FIGARO, authors' calculations.

Table 3 shows that the shock raises average euro-area inflation by 5.7 percentage points in the 2022 simulation. Household energy contributes 2.2 points, while fuel and transport excluding fuel contribute 1.0 and 0.7 points. Other products and food contribute 0.7 and 0.5 point. These product contributions combine the sectoral price response from the 2022 FIGARO network with each product's 2022 weight in the relevant household basket. The 10 percent food shock in panel (b) raises average inflation by 1.0 point but changes the Q1–Q5 gap by only 0.02 point before rounding.

The distributional effect is concentrated in household energy. The shock raises Q1 inflation through water, electricity, gas, and other fuels by 2.2 points, compared with 1.5 points for Q5, adding 0.7 point to the Q1–Q5 gap. Higher-income households are more exposed to motor fuels and transport; these two categories lower the gap by 0.5 and 0.1 point. The total simulated gap is 0.2 point, split almost equally between direct exposure and indirect production-network prop-

agation. The simulated gap and the cumulative gap in Table 2 differ in sign and should not be interpreted as a model fit: the simulation applies a uniform shock with complete pass-through, whereas observed consumer prices also reflect country-specific taxes, subsidies, regulated tariffs, and other price shocks.

The energy shock widens the income gap mainly because lower-income households devote more of their budgets to electricity, gas, and other household fuels. Production networks reinforce this direct exposure, but account for less than one third of the simulated gap. Price caps, tariff shields, and tax or charge reductions can therefore reduce inflation inequality most strongly when they lower the retail prices of household energy, the component that generates the largest direct gap.

3 Effects of Household Fiscal Policies on Inflation Inequality

This section evaluates the distributional effects of unconventional fiscal policies implemented during the 2021–2023 energy crisis that directly changed consumer prices faced by households, including electricity and gas price caps and shields, VAT and energy-tax reductions, cuts in regulated charges, and fuel rebates.

3.1 Constructing No-Policy Counterfactual Price Indices

In response to the energy crisis, euro-area governments introduced a broad array of measures to mitigate household exposure to rising energy costs (Arregui et al., 2022, Dao et al., 2023, García-Miralles, 2023, Levell et al., 2026). These included both direct income support, such as cash transfers to vulnerable groups, and price-based interventions, including temporary VAT reductions and price ceilings on gas, electricity, and fuel. Despite this variety, governments in the euro area relied predominantly on measures that directly reduced consumer energy prices rather than on narrowly targeted income support. Combining the Bruegel dataset with national sources for Spanish and Portuguese food-product VAT cuts, measures acting directly on prices in the 20 euro-area countries account for EUR 268.0 billion. Of this price-based support, EUR 235.2 billion is delivered through untargeted measures and EUR 32.8 billion through measures targeted at vulnerable households.

The measures differ substantially across euro-area countries. Some governments rely primarily on regulated retail-price caps or tariff shields, while others used VAT and excise-tax reductions, fuel rebates, system-charge reductions, or calibrated block-tariff schemes. Table B.1 in the

appendix reports the euro-area inventory and identifies the measures simulated in our price counterfactuals. We construct these indices for measures for which the necessary statutory, price, and unit-conversion information is available; Appendix B reports the country-level series.

We compare observed policy-inclusive retail prices with estimated no-policy price paths for the affected products. Because the measures operate through administered prices, tax-inclusive prices, retail price ceilings, or unit rebates, their direct effects are confined to the covered product categories.

We distinguish three cases according to the information available. Measures with observed no-policy tariffs, statutory tax rates, or statutory unit rebates require no calibration. When the inventory lacks the monthly tariff path, unit price, or coverage weights needed for direct reconstruction, we choose one measure-specific intensity parameter so that the mean of the implied monthly no-policy price ratios over the active policy months equals the aggregate-equivalent ratio.

Counterfactual price indices are constructed at COICOP digit 3 by replacing each observed HICP component index with its no-policy counterpart. Let $P_{j,t}$ denote the observed HICP component index for product j in month t , and let $\tilde{P}_{j,t}$ denote the corresponding counterfactual index. For each simulated measure, we recover a counterfactual retail price $\tilde{p}_{j,t}$ and apply the implied price ratio to the observed component index:

$$\tilde{P}_{j,t} = P_{j,t} \frac{\tilde{p}_{j,t}}{p_{j,t}}. \quad (12)$$

VAT and Excise-Tax Reductions. For VAT and excise changes, we back out the reduced tax-inclusive price and reapply the normal rate; coverage and pass-through are calibrated only when full statutory coverage cannot be established. For VAT or excise-tax reductions, the observed retail price includes the reduced policy tax rate. Following the standard tax-inclusive price logic used to assess the inflation effects of VAT changes (Gautier and Lalliard, 2013), the no-policy price is obtained by backing out the reduced tax-inclusive price and reapplying the normal tax rate:

$$\tilde{p}_{j,t} = p_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad (13)$$

where $\tau_{j,t}^1$ is the policy tax rate and $\tau_{j,t}^0$ is the no-policy tax rate.

Administered Prices. For regulated-price shields, price caps, and administered tariffs, we use an observed no-policy reference tariff when available and otherwise calibrate an aggregate-equivalent price wedge.

For regulated or administered-price measures, such as electricity or gas tariff shields and retail price caps, the counterfactual price is taken from a no-policy reference path:

$$\tilde{p}_{j,t} = p_{j,t}^{\text{ref}}. \quad (14)$$

When the observed tariff and the no-policy reference tariff are both available, the no-policy retail price is obtained by rescaling the observed price. If $T_{j,t}^{\text{obs}}$ denotes the regulated tariff actually applied and $T_{j,t}^{\text{ref}}$ the tariff that would have applied without the intervention, then:

$$p_{j,t}^{\text{ref}} = p_{j,t} \frac{T_{j,t}^{\text{ref}}}{T_{j,t}^{\text{obs}}}. \quad (15)$$

Subsidies and Rebates. For unit subsidies and fuel rebates, we add the statutory absolute price wedge to the observed monthly price path or, without a retail unit-price series, calibrate it in HICP component-index units. Evidence from [Drolsbach et al. \(2023\)](#) suggests that temporary fuel-tax reductions and discounts were largely passed through to consumer prices in 2022 in several euro-area countries, although pass-through varied over time and by fuel type. This supports treating these measures as retail-price wedges in the counterfactual construction. For unit subsidies, fuel rebates, or per-kWh support schemes, the observed price is lower than the no-policy price by the statutory support amount. The counterfactual price is obtained by adding the support amount $s_{j,t}$ back to the observed unit price:

$$\tilde{p}_{j,t} = p_{j,t} + s_{j,t}. \quad (16)$$

For transport-fuel rebates, $s_{j,t}$ is recovered from the statutory pump discount, expressed in euros per litre, and applied to a monthly retail-fuel price consistent with the HICP transport-fuel component. Following [Bonnet et al. \(2025\)](#), we refer to these interventions as pump-price rebates or subsidies because they reduced the tax-inclusive price paid by consumers, although their technical implementation sometimes operated through fuel excise taxation rather than through a direct transfer to households. The no-rebate fuel index is obtained by rescaling the observed HICP transport-fuel component:

$$\tilde{P}_{j,t} = P_{j,t} \frac{p_{j,t}^M + s_{j,t}}{p_{j,t}^M}.$$

More generally, when the Bruegel dataset identifies a subsidy but does not supply a monthly unit price, we set $s_{k,t} = \delta_k \bar{P}_j$ in component-index units. This yields the time-varying ratio $1 + \delta_k \bar{P}_j / P_{j,t}$. The single parameter δ_k is calibrated.

For a small set of price-relevant interventions, the policy inventories identify the measure but do not provide the monthly no-policy tariff schedule, tariff-block utilization, or eligible-household

weights needed for a direct statutory reconstruction. We therefore add these cases as separate calibrated layers, identified from [Sgaravatti et al. \(2021\)](#) and [Arregui et al. \(2022\)](#), rather than merging them with the VAT or unit-subsidy calculations. This applies to block-tariff or price-cap schemes in Germany, the Netherlands, Cyprus, Ireland, Italy, and Slovenia, and to Belgium’s extension of the social energy tariff and basic energy package. The calibrated ratios are applied only to the relevant COICOP energy components and policy months, and the corresponding assumption tables are reported with the counterfactual outputs. These layers should be interpreted as aggregate-equivalent price effects, not as statutory tariff reconstructions.

Table 4 summarizes the resulting policy decomposition. Administered-price measures account for most of the reduction in the Q1–Q5 inflation gap (−0.9 point). Fuel subsidies and rebates have a small cumulative effect because their positive effect in 2022 is largely reversed in 2023. The targeted measures account for −0.5 point of the total −1.1-point reduction despite representing EUR 32.8 billion of the EUR 268.0 billion in price support.

Table 4: Contributions of household support categories to average inflation and inflation inequality in the euro area, January 2021–December 2023

Policy category	Fiscal cost (EUR bn)	Average inflation				Q1–Q5 inflation gap				Age inflation gap		Residence-area gap
		2021–2023	2021	2022	2023	2021–2023	2021	2022	2023	2021–2023	2021–2023	
Instrument classification												
VAT cuts	47.8	-0.4	-0.1	-0.2	-0.1	-0.2	0.0	-0.1	-0.1	-0.2	-0.1	
Administered prices	173.0	-1.4	0.0	-0.5	-0.8	-0.9	0.0	-0.3	-0.5	-0.7	-0.5	
Fuel subsidies and rebates	20.1	0.0	0.0	-0.3	0.3	0.0	0.0	0.1	-0.1	0.0	0.0	
Other calibrated layer	27.1	0.0	0.0	-0.1	0.1	-0.1	0.0	-0.1	0.0	0.0	0.0	
Total	268.0	-1.9	-0.1	-1.2	-0.5	-1.1	-0.1	-0.4	-0.5	-0.9	-0.6	
Targeted vs Non-targeted												
Untargeted price measures	235.2	-0.8	0.0	-0.7	0.0	-0.6	0.0	-0.2	-0.3	-0.3	-0.2	
Targeted price measures	32.8	-1.1	0.0	-0.5	-0.5	-0.5	0.0	-0.3	-0.2	-0.6	-0.4	
Total	268.0	-1.9	-0.1	-1.2	-0.5	-1.1	-0.1	-0.4	-0.5	-0.9	-0.6	

Notes. Negative values indicate that policies reduced inflation or the gap. Fiscal costs are in EUR billions. Each row removes one policy family; Total removes all policies jointly and may differ from the row sum because of interactions. No residual is allocated. Sources: Bruegel, national sources, authors’ calculations.

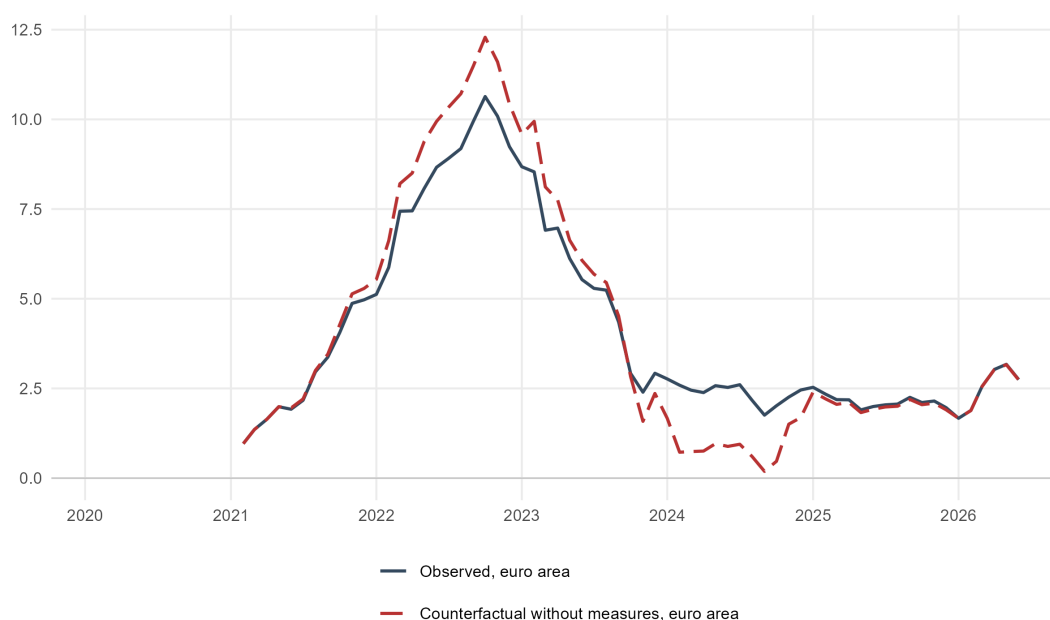
3.2 Effects of Unconventional Fiscal Policies

The price-based measures changed both the level and the timing of inflation during the energy crisis. We compare observed inflation with counterfactual paths that remove fiscal policies. This comparison isolates their direct effect on consumer prices.

Figure 6 shows the effect of unconventional policies on average euro-area inflation. The no-

policy index lies above observed inflation during the main phase of the energy shock, with the largest difference around the inflation peak in late 2022. VAT and excise reductions, fuel rebates, price shields, and regulated-price caps therefore compressed the peak rather than preventing the underlying rise in inflation. As measures expire, the ordering temporarily reverses: year-on-year inflation is then calculated relative to policy-depressed prices twelve months earlier. This base effect shifts measured inflation across time and should not be interpreted as evidence that the measures raised the underlying price level.

Figure 6: Household price policies: observed and no-policy euro-area inflation (year on year, in %)



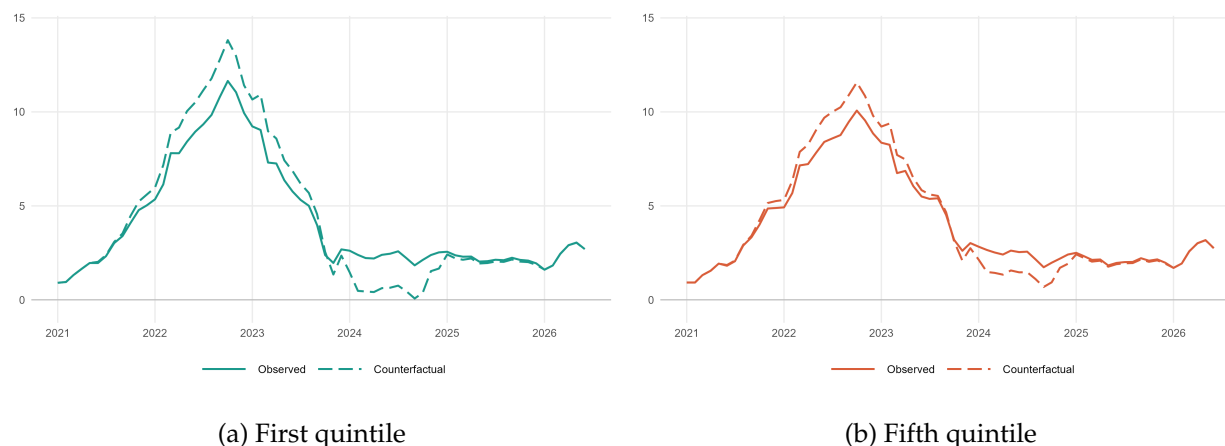
Notes: The no-policy series removes fiscal interventions that directly affected consumer prices. The figure reports year-on-year inflation for all 20 euro-area countries, aggregated with annual HICP country weights, from January 2021 to July 2026. Cumulative policy effects reported in the text cover 2021–2023 and are defined as observed minus no-policy inflation.

Sources: Eurostat HICP; Bruegel energy-policy dataset; authors' calculations.

Figure 7 shows that, without the price measures, inflation for the first income quintile would have exceeded that of the fifth quintile by a wider margin during the 2022 peak. The gap subsequently narrows and reverses during much of 2024 as energy inflation falls and the earlier policy wedges enter the year-on-year comparison base. Together with the observed paths in Figure 3, this pattern indicates that the measures primarily protected lower-income households when energy prices were rising fastest. This finding is consistent with microsimulation evidence that the

European fiscal response reduced distributional losses, although broad price measures were less targeted than income support (Amores et al., 2025, Pallotti et al., 2024).

Figure 7: Household price policies: no-policy euro-area inflation by income quintile (year on year, in %)



Notes: The figure reports year-on-year inflation rates in the absence of household price policies for the first and fifth income quintiles, from January 2021 to July 2026. Income transfers and firm-only measures are outside the counterfactual. All 20 countries, including Italy and Spain, use RAS expenditure weights and are aggregated using annual HICP country weights.

Sources: Eurostat HICP and country weights; household budget surveys; Bruegel energy-policy dataset; authors' calculations.

3.3 Heterogeneity Across Countries and Products

We first compare countries and then decompose the euro-area effect by product group.

3.3.1 Differences Across Countries

Table 5 reports the fiscal cost of household price policies and cumulative no-policy inflation, together with the corresponding income-quintile indices and inflation gap. The fiscal costs provide context for the scale of the interventions⁶.

⁶These fiscal costs are not inputs into the price-index counterfactual and do not measure the monetary benefit received by each household group

Table 5: Household price policies: fiscal cost and no-policy inflation in the 20 euro-area countries, January 2021–December 2023

Country	Weight (2022, %)	Fiscal cost (% of 2022 GDP)	Counterfactual inflation	Effect of price policies on average inflation	Counterfactual inflation gap	Effect of price policies on inflation inequality
Germany	28.1	1.5	19.4	-0.9	1.4	-0.9
France	20.4	3.0	19.1	-4.5	3.0	-2.3
Italy	16.3	1.8	19.3	-0.7	1.6	-0.3
Spain	11.0	1.7	19.3	-2.7	1.2	-1.4
Netherlands	5.4	3.5	22.4	-2.4	1.8	-1.2
Belgium	4.0	0.9	19.4	-0.3	0.3	-0.6
Austria	3.3	0.9	23.3	-1.3	0.3	-0.7
Portugal	2.2	2.4	18.0	-2.6	1.0	-1.4
Greece	2.2	4.0	19.1	-1.2	2.8	-1.4
Finland	1.9	0.2	13.3	-0.1	-1.1	0.1
Ireland	1.5	0.3	17.8	0.0	4.4	-0.2
Slovakia	0.8	2.1	31.8	-3.3	4.3	-2.2
Croatia	0.7	1.3	25.3	-0.6	0.1	-0.5
Lithuania	0.6	1.6	34.3	0.0	3.9	-0.3
Slovenia	0.4	0.3	21.3	-0.1	1.5	-0.2
Luxembourg	0.3	1.2	15.5	-1.0	0.6	-0.4
Latvia	0.3	1.9	30.8	0.0	6.7	-0.8
Estonia	0.2	0.5	36.6	-0.1	6.9	-0.1
Cyprus	0.2	0.7	15.9	0.0	0.3	0.0
Malta	0.1	4.5	17.2	-1.5	1.9	-0.2
Euro area	100.0	1.9	19.7	-1.9	1.8	-1.1

Notes. The no-policy counterfactual removes household price policies. Effects are observed minus no-policy values; the gap is Q1 minus Q5. Cumulative changes compare January 2021 with December 2023. Fiscal costs are percentages of 2022 nominal GDP; weights are 2022 EA20 HICP shares. Sources: Eurostat, national statistical institutes, Bruegel, national sources, authors' calculations.

The distributional effects are heterogeneous across countries. Among the countries included in the euro-area aggregate, the largest reductions in the Quintile 1–Quintile 5 gap are estimated for France (–2.3 points), Slovakia (–2.2), Spain, Portugal, and Greece (–1.4 each), and the Netherlands (–1.2). Germany has an effect of –0.9 point and Italy –0.3 point. Estimates close to zero arise either where no policy layer from Appendix B enters the country row or where temporary counterfactual price wedges leave little difference in the cumulative 2021–2023 comparison.

France combines a reported budget of EUR 80.5 billion with a 2.3-point reduction in the Q1–Q5 inflation gap, whereas Slovakia combines EUR 2.3 billion with a 2.2-point reduction. These

figures refer to economies of different sizes, policy coverage, energy systems, and initial price exposures. The inflation-gap effect is measured in percentage points, not as a monetary gain, and it cannot be divided meaningfully by expenditure in euros. A cost-effectiveness analysis would need household-level policy eligibility, quantities consumed, tax and transfer rules, and counterfactual expenditures to calculate the fiscal cost and purchasing-power benefit for each household in a common monetary unit.

3.3.2 Differences Across Products

For the euro area, cumulative no-policy inflation is 19.7 percent over 2021–2023, compared with 17.8 percent in the observed index. The observed Quintile 1–Quintile 5 gap is 0.6 point, compared with 1.8 points without the simulated measures. Using unrounded indices, household price policies therefore lower average inflation by 1.9 points and the income gap by 1.1 points.

Table 6 reproduces the product-group decomposition of Table 2 using the no-policy paths. It reports each product group’s contribution to cumulative inflation and to the Quintile 1–Quintile 5 gap, together with the annual and cumulative effects of the price measures. The observed and counterfactual decompositions use the same product groups, weights, and aggregation rules.

Table 6: Annual product-group contributions of household price policies to the euro-area Q1–Q5 inflation gap

Product group	Weight (2022, %)	Counterfactual inflation	Effect of policies on average inflation	Counterfactual inflation gap	Effect of price policies on inflation inequality			
					2021– 2023	2021	2022	2023
Food and non-alcoholic beverages	16.6	4.7	-0.5	1.5	-0.1	0.0	0.0	0.0
Alcoholic beverage, tobacco and narcotics	4.3	0.9	-0.1	0.4	0.0	0.0	0.0	0.0
Clothing and footwear	5.2	0.9	-0.1	-0.3	0.0	0.0	0.0	0.0
Housing (rentals and repairs)	12.2	2.3	-0.2	-0.4	0.0	0.0	0.0	0.0
Water, electricity, gas and other fuels	6.3	3.7	-0.3	3.0	-1.1	-0.1	-0.5	-0.5
Fuel	5.9	1.7	-0.2	-0.5	0.0	0.0	0.1	0.0
Transport (excluding fuel)	8.9	2.1	-0.2	-0.6	0.0	0.0	0.0	0.0
Other	40.5	3.4	-0.3	-1.3	0.0	0.0	0.0	0.0
Total	100.0	19.7	-1.9	1.8	-1.1	-0.1	-0.4	-0.5

Notes. Policy effects are observed minus no-policy contributions; the gap is Q1 minus Q5. Annual columns compare annual-average indices. Cumulative columns use the same chain-linked January 2021–December 2023 indices as the country table. Weights are 2022 EA20 expenditure shares. Totals are exact before rounding; no normalization or residual is used. Sources: Eurostat, national statistical institutes, authors’ calculations.

Household energy accounts for the entire displayed cumulative reduction in the income gap

(−1.1 points). Its annual contribution is −0.1 point in 2021 and −0.5 point in both 2022 and 2023. The total annual effects across all products are −0.1, −0.4, and −0.5 point, respectively.

4 Household-Level Inflation

This section moves from inflation indices for broad household groups to estimates constructed from individual household records. We combine each household’s consumption basket, reported in the HBS microdata, with monthly HICP price indices. This approach shows the dispersion within groups and allows us to examine whether observable household characteristics are systematically associated with inflation exposure.

4.1 Calculating Household Inflation Rates

We apply fixed-basket Laspeyres at the household level, holding each household’s 2020 HBS basket fixed when calculating inflation for 2021–2023. Let $e_{i,j,2020}^{\text{HBS}}$ denote household i ’s expenditure on COICOP item j in the 2020 HBS wave, and let $a_{i,2020}$ denote its survey weight. The household expenditure share is

$$\omega_{j,i,2020}^{\text{HBS}} = \frac{a_{i,2020} e_{i,j,2020}^{\text{HBS}}}{\sum_{k \in \mathcal{J}_{i,2020}} a_{i,2020} e_{i,k,2020}^{\text{HBS}}} = \frac{e_{i,j,2020}^{\text{HBS}}}{\sum_{k \in \mathcal{J}_{i,2020}} e_{i,k,2020}^{\text{HBS}}} \quad (17)$$

using the same convention as the group-specific weights $\omega_{j,g,y}$ defined above. Since the same survey weight applies to all items within a household, it cancels out in the within-household basket; it is then used to weight households when constructing cross-household distributions and summary statistics. For household i , the within-year price index in month m of year y is

$$P_{i,y,m} = \sum_{j \in \mathcal{J}_{i,2020,y}} \omega_{j,i,2020}^{\text{HBS}} \frac{p_{j,y,m}}{p_{j,y,0}}, \quad (18)$$

where $\mathcal{J}_{i,2020,y}$ is the set of household expenditure items observed in the 2020 HBS wave that can be matched to HICP prices in year y . Households are therefore treated as elementary household groups: the within-year Laspeyres changes are chain-linked at the household level, and year-on-year monthly inflation rates $\pi_{i,y,m}$ are computed from the resulting chained household price index.

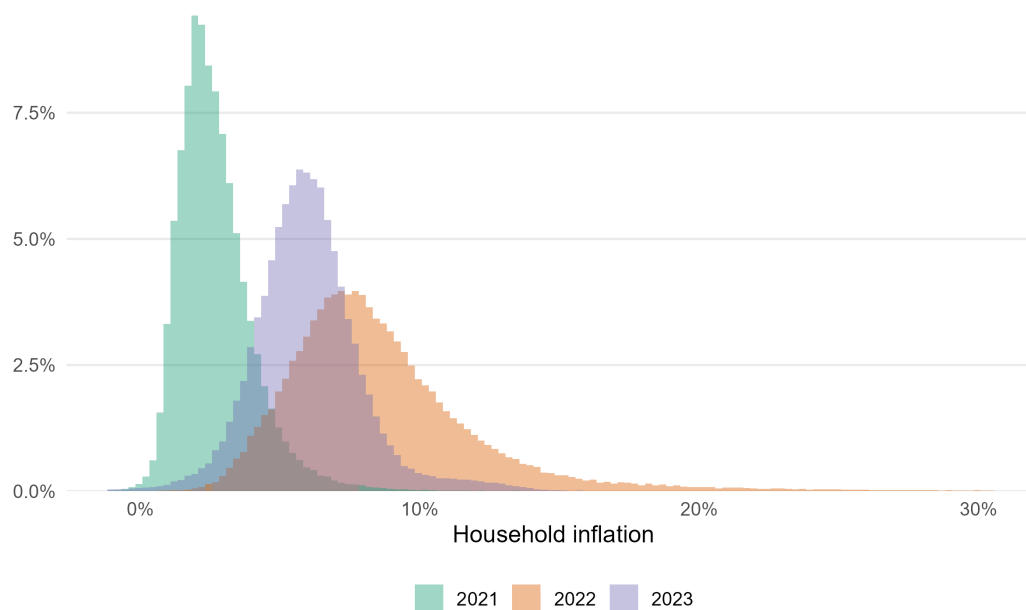
For each household i in country c and calendar year y , average annual inflation is the arithmetic mean of the 12 monthly year-on-year inflation rates. If $I_{ic,y,m}$ is the chained household price index, then

$$\pi_{ic,y,m} = 100 \left(\frac{I_{ic,y,m}}{I_{ic,y-1,m}} - 1 \right), \quad \bar{\pi}_{ic,y} = \frac{1}{12} \sum_{m=1}^{12} \pi_{ic,y,m}. \quad (19)$$

This annual average is the inflation measure used in the household-level regressions and in the inflation-burden calculation below.

Figure 8 shows the dispersion of these household-level inflation rates across the euro-area microdata sample in 2021, 2022, and 2023.

Figure 8: Distribution of household-level inflation in the euro area, 2021–2023 (in % of households)



Notes: Histograms report the survey-weighted percentage of households by annual inflation interval, calculated from fixed 2020 household consumption baskets. The displayed range excludes the outer 0.1 percent of the pooled weighted distribution. The sample covers 19 euro-area countries; Portugal is excluded because no household basket can be matched to the annual HICP series.

Sources: Eurostat HICP and HBS microdata; authors' calculations.

4.2 Inflation Burden

Our indicators measure differences in inflation exposure across household groups, but they do not directly express the cost of inflation relative to household resources. We therefore construct the burden from HBS microdata rather than combining group-level inflation with an external aggregate consumption-to-income ratio. For household i in country c , let X_{ic} denote total consumption expenditure, Y_{ic} net household income, and $\bar{\pi}_{ic,y}$ the household's average annual inflation rate in year y , computed from its own COICOP expenditure shares and the corresponding national HICP

price changes. The raw household-level burden is

$$b_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}, \quad (20)$$

where $\bar{\pi}_{ic,y}$ is measured in percent, so $b_{ic,y}^{\text{raw}}$ is measured in points of net income. Thus, the indicator allows both individual inflation exposure and the consumption-to-income ratio to vary across households.

Very small reported incomes can generate mechanically extreme expenditure-to-income ratios. We therefore winsorize $100X_{ic}/Y_{ic}$ at the survey-weighted 99th percentile within each country, while retaining the underlying household inflation rate. Denoting the resulting ratio by $\tilde{\kappa}_{ic}$, the burden used in the estimates and its aggregation for national income quintile q are

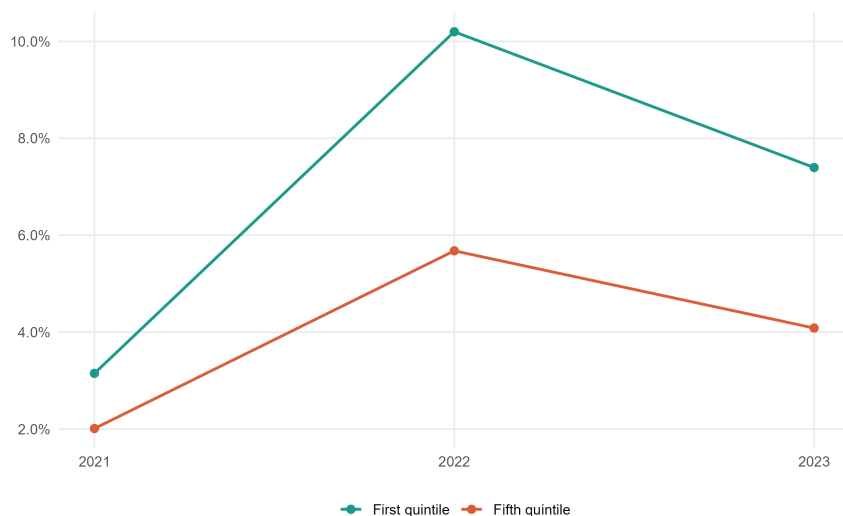
$$b_{ic,y} = \bar{\pi}_{ic,y} \frac{\tilde{\kappa}_{ic}}{100}, \quad B_{q,y} = \frac{\sum_c \sum_{i \in q(c)} a_{ic} b_{ic,y}}{\sum_c \sum_{i \in q(c)} a_{ic}}, \quad (21)$$

where a_{ic} is the HBS survey weight and $q(c)$ indicates that equivalized-income quintiles are defined separately within each country before households are pooled. The survey weights expand the microdata to national household populations and therefore determine the euro-area aggregation. The 2020 HBS basket, income, and expenditure observations are held fixed, while household-specific inflation is calculated for 2021–2023. For each household-year, the inflation rate is computed over the expenditure categories matched to available HICP items and the matched shares are renormalized to one. We impose no minimum matched-expenditure threshold.

Higher-income households also generally have greater capacity to absorb a price shock by reducing discretionary expenditure or substituting toward cheaper varieties. Appendix Figure A.2 documents the corresponding difference in basket composition.

Figure 9 reports the weighted household-level burden. The first quintile faces a substantially larger burden in every year: 3.1 percent of net income in 2021, 10.2 percent in 2022, and 7.4 percent in 2023, compared with 2.0, 5.7, and 4.1 percent for the fifth quintile. The Q1–Q5 burden gap therefore widened from 1.1 points in 2021 to 4.5 points at the peak of the energy-price shock in 2022, before narrowing to 3.3 points in 2023.

Figure 9: Household-level inflation burden in the euro area, by income quintile (in % of net income)



Notes: The burden is computed for each household as its annual basket-specific inflation rate multiplied by its consumption-to-net-income ratio. Ratios are winsorized at the survey-weighted national 99th percentile; income quintiles are defined within countries. Estimates use the 2020 HBS microdata and annual inflation for 2021–2023. Italy is excluded because the harmonized micro-income input is unavailable. Portugal is absent because no household basket can be matched to the annual HICP series. The resulting sample covers 18 euro-area countries.

Sources: Eurostat HICP and HBS microdata; authors' calculations.

4.3 Household Characteristics and Inflation Exposure

We pool individual household records from the euro area to examine whether differences in consumption baskets translate into systematic differences in inflation exposure. We examine how income quintile, age, household size, and degree of urbanization are associated with household-specific inflation.

Table 7 reports separate regressions for 2021, 2022, and 2023, with country fixed effects. Table 8 pools the three years and includes both country and year fixed effects. Hence, the coefficients capture conditional differences between households within countries, net of the included characteristics and common country effects; the pooled estimates additionally net out the large common changes in inflation across years. They should be interpreted as descriptive conditional associations, not as causal effects.

Table 7: Household-level inflation regressions by year, euro-area sample

Dependent Variable:	mean_inflation		
Year	2021	2022	2023
Model:	(1)	(2)	(3)
<i>Variables</i>			
Less than 30 years	-0.1692 (0.1129)	-2.005*** (0.2119)	-0.6680** (0.2426)
From 30 to 44 years	-0.0156 (0.0844)	-1.391*** (0.1612)	-0.5693*** (0.0890)
From 45 to 59 years	0.0140 (0.0892)	-0.8904*** (0.1706)	-0.2963*** (0.0771)
Income quintile 2	0.1113* (0.0602)	-0.6340** (0.2379)	-0.1970** (0.0737)
Income quintile 3	0.1693 (0.1142)	-0.6450** (0.2551)	-0.1542 (0.0905)
Income quintile 4	0.2371* (0.1362)	-0.6881*** (0.2367)	-0.1698* (0.0962)
Income quintile 5	0.3104* (0.1562)	-0.8511*** (0.2391)	-0.2247** (0.1008)
Household size	0.0209 (0.0415)	-0.0696 (0.0800)	0.1153*** (0.0389)
Intermediate area	-0.2723*** (0.0809)	-0.7404*** (0.1223)	0.0201 (0.0456)
Densely populated area	-0.6118*** (0.0979)	-1.432*** (0.1364)	-0.0360 (0.1197)
<i>Fixed-effects</i>			
country	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	152,438	152,440	152,440
R ²	0.31975	0.66935	0.63879
Within R ²	0.04698	0.09037	0.02904
<i>Clustered (country) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Notes: The dependent variable is mean annual household inflation, in percent. Regressions are estimated separately by inflation year using household survey weights and include country fixed effects. The omitted categories are income quintile 1 and rural areas. The estimation sample covers 18 euro-area countries; Italy lacks usable harmonized household income and Portugal has no household basket that can be matched to the annual HICP series. Standard errors clustered by country are reported in parentheses.

Sources: Household Budget Survey microdata; Eurostat HICP; authors' calculations.

All three regressions use household records from the same 2020 HBS wave and impose no minimum matched-expenditure threshold. Annual inflation can nevertheless be calculated only when at least one category in a household's basket matches an HICP item with sufficient monthly

price coverage. After removing the matching threshold, the estimation sample contains 152,438 observations in 2021 and 152,440 in both 2022 and 2023. Only two regression observations therefore lack sufficient price matches in 2021. The effective estimation sample is otherwise constant across years.

Table 8: Pooled household-level inflation regression, euro-area sample

Dependent Variable:	mean_inflation
Model	Pooled
Model:	(1)
<i>Variables</i>	
Less than 30 years	-0.9473*** (0.1085)
From 30 to 44 years	-0.6585*** (0.0708)
From 45 to 59 years	-0.3909*** (0.0813)
Income quintile 2	-0.2398** (0.1024)
Income quintile 3	-0.2099 (0.1273)
Income quintile 4	-0.2069 (0.1269)
Income quintile 5	-0.2551* (0.1327)
Household size	0.0222 (0.0457)
Intermediate area	-0.3309*** (0.0521)
Densely populated area	-0.6933*** (0.0723)
<i>Fixed-effects</i>	
country	Yes
year	Yes
<i>Fit statistics</i>	
Observations	457,318
R ²	0.62435
Within R ²	0.02438
<i>Clustered (country) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Notes: The dependent variable is mean annual household inflation, in percent. The regression uses household survey weights and includes country and inflation-year fixed effects. The omitted categories are income quintile 1 and rural areas. The estimation sample covers 18 euro-area countries; Italy lacks usable harmonized household income and Portugal has no household basket that can be matched to the annual HICP series. Standard errors clustered by country are reported in parentheses.

Sources: Household Budget Survey microdata; Eurostat HICP; authors' calculations.

The income-quintile coefficients change sign over time. Relative to the first quintile, households in the upper quintiles experience higher inflation in 2021: the coefficients rise from 0.11 point for the second quintile to 0.31 point for the fifth. In 2022, all four coefficients are negative

and statistically significant, ranging from -0.63 to -0.85 points, with the largest difference for the fifth quintile. The coefficients remain negative in 2023, between -0.15 and -0.22 points; three of the four are statistically different from zero at the 10 percent level. In the pooled specification, they range from -0.21 to -0.26 point. The second-quintile coefficient is significant at 5 percent and the fifth-quintile coefficient at 10 percent, whereas the middle-quintile coefficients are not statistically different from zero. Thus, higher-income households have slightly higher inflation in 2021 but lower inflation during 2022–2023; the results do not support a stable linear relationship between income rank and household inflation over the full period.

Age differences are largest during the energy-price shock. Relative to households whose reference person is aged 60 or over, households under 30 experience 2.01 points lower inflation in 2022 and 0.67 point lower inflation in 2023. The corresponding differences for households aged 30–44 are -1.39 and -0.57 points, and those for households aged 45–59 are -0.89 and -0.30 points. The pooled coefficients preserve this age gradient, ranging from -0.40 point for ages 45–59 to -0.95 point for households under 30.

Household size is also associated with inflation exposure, but its sign varies across years. Each additional household member is associated with 0.02 point higher inflation in 2021, 0.07 point lower inflation in 2022, and 0.12 point higher inflation in 2023. The pooled estimate is positive but small, at 0.02 point. This instability suggests that household size mainly proxies for differences in baskets whose inflation sensitivity depends on the composition of the shock, rather than a persistent exposure gradient.

Residence differences are concentrated in 2021–2022. Conditional on the other covariates and country effects, inflation in 2022 is estimated to be 0.74 point lower in intermediate-density areas and 1.43 points lower in densely populated areas than in rural areas. In 2023, the differences narrow to 0.02 point higher inflation in intermediate areas and 0.04 point lower inflation in densely populated areas. The pooled estimates are respectively -0.33 and -0.69 point. The large 2022 gaps are consistent with rural households allocating larger shares to home energy and private transport, the categories most exposed to the energy-price shock.

5 Conclusion

This paper constructs indicators to measure inflation inequalities across socio-demographic groups in the euro area. Over 2021–2023, the cumulative Quintile 1–Quintile 5 gap is 0.6 point, compared with 1.3 points between households aged 60 or over and those under 30, and 1.0 point between

rural and urban households. These differentials are concentrated around the energy-price surge and narrow as energy inflation recedes. Inflation inequalities remain limited in a low-inflation environment.

Similar average inflation rates nevertheless imposed sharply different burdens. In 2022, inflation absorbed 10.2 percent of net income for households in the first income quintile, compared with 5.7 percent for those in the fifth. Household characteristics also mattered. Age and residence show the largest conditional differences during the energy shock: the rural–urban gradient is pronounced in 2021–2022 but vanishes in 2023, while the age gradient persists through 2023. Income and household size display less stable relationships across years. VAT reductions, fuel rebates, and energy price shields lower average cumulative inflation by 1.9 points and reduce the aggregate Quintile 1–Quintile 5 gap by 1.1 points in the direct no-policy comparison.

In the policy counterfactual, the direct index comparison shows that the measures reduce the Q1–Q5 gap by 1.1 points. The product decomposition assigns this reduction to household energy. The measures lower the retail energy prices to which lower-income households are directly exposed. These measures have little effect on price increases transmitted through firms’ input costs.

Income alone does not identify vulnerable households: residence and age also capture differences in heating technology, mobility constraints, and the ability to substitute between products. More granular transaction and expenditure data would allow future work to measure substitution within product categories.

An extension of this article would combine household-specific price indices with harmonized euro-area household balance-sheet surveys to assess the heterogeneous effects of inflation on household wealth, particularly across debtors and creditors and across households with different asset and liability compositions.

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Appendix

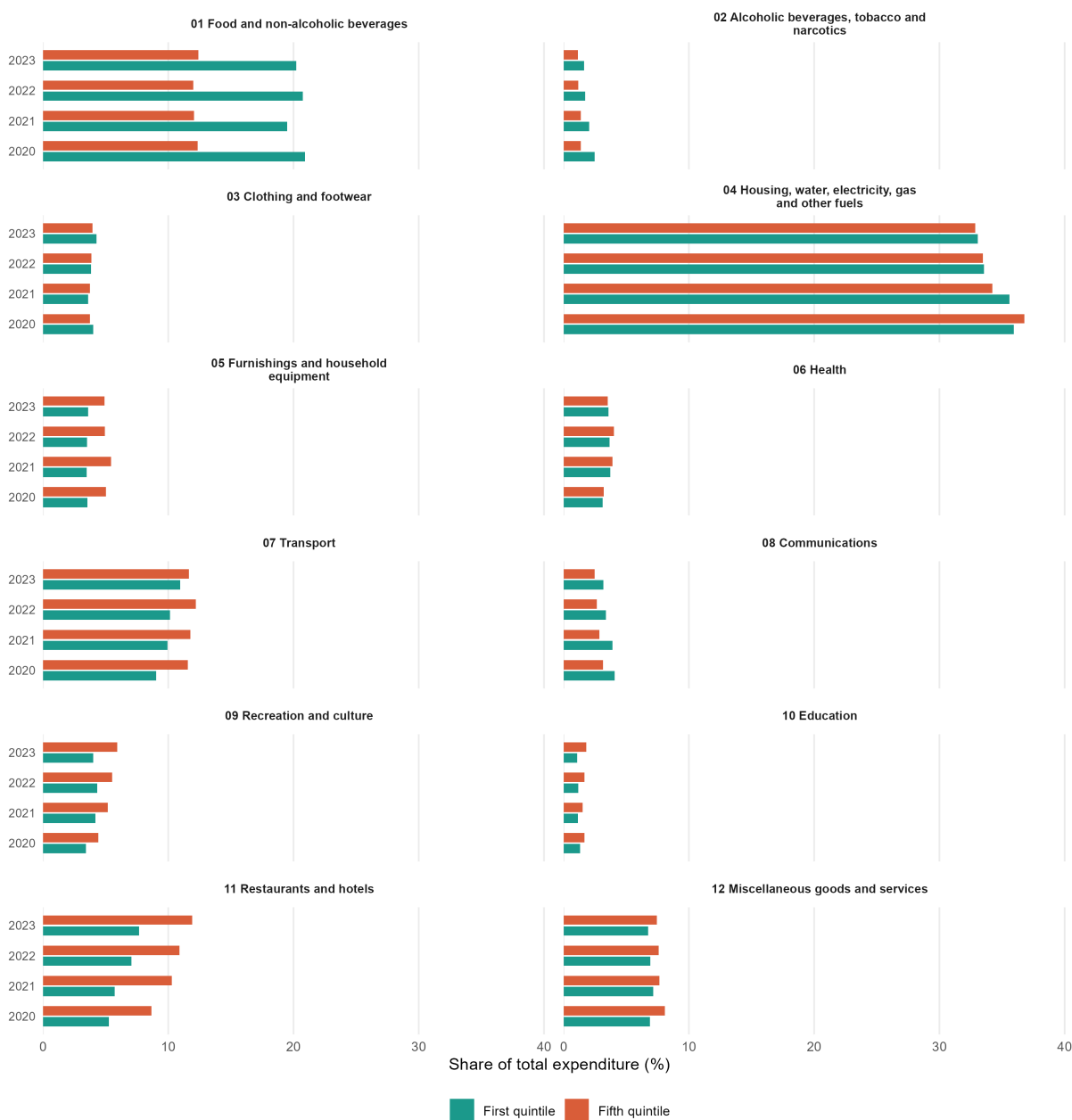
Erwan Gautier - Jérémie Montornès

(Not for publication)

August 31, 2026

A Additional Figures on the Household Budget Surveys

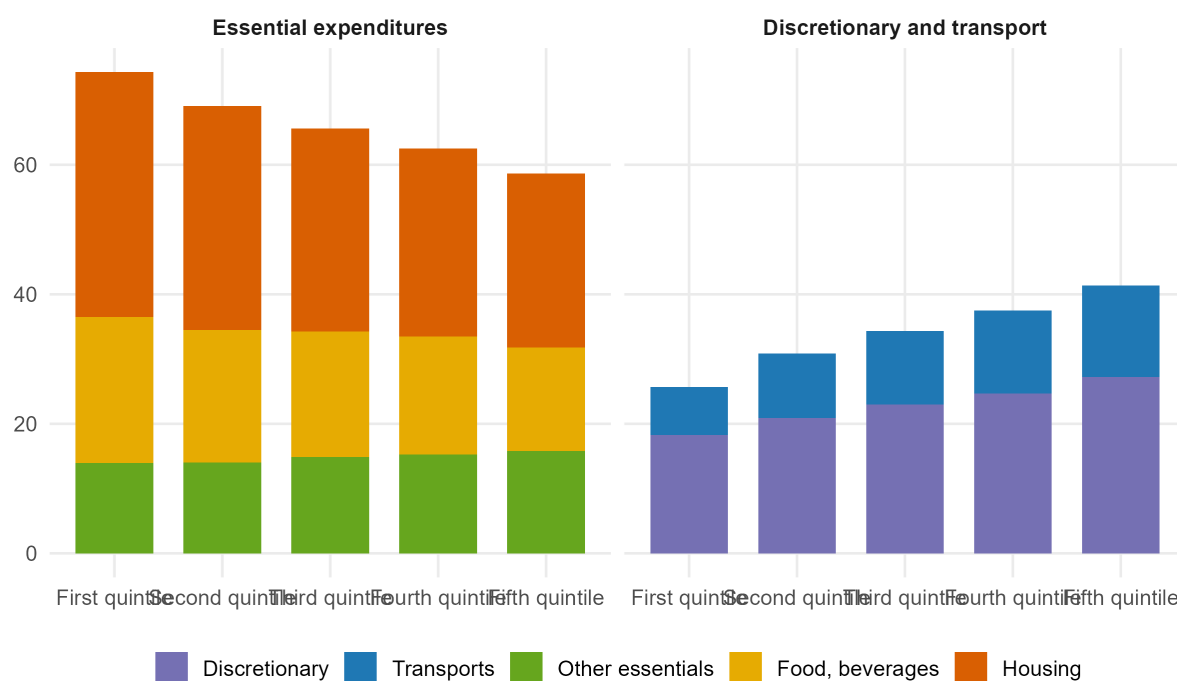
Figure A.1: Main COICOP expenditure weights by income quintile, Spain (in % of total expenditure)



Notes: The figure reports the share of each main COICOP category in total household expenditure for the first and fifth equivalized-income quintiles. Quintiles are computed separately by year using household weights.

Sources: INE Encuesta de Presupuestos Familiares public-use microdata; authors' calculations.

Figure A.2: Essential and discretionary expenditures by income quintile, euro area (in % of total expenditure)



Notes: The figure partitions COICOP expenditure shares into essential expenditures and discretionary/transport expenditures. The euro-area aggregate is computed as the unweighted average of available country-level HBS expenditure shares.

Sources: Eurostat HBS 2015; authors' calculations.

B Counterfactuals

This appendix reports country-level observed and counterfactual headline inflation series. Each figure aggregates all simulated energy and food price-policy layers available for the country, using the HICP item weights in the country CPI basket.

Table B.1: Price measures included in the national counterfactual simulations

Country	Included price measures	COICOP nents	compo- Assumption window	Input status
Austria	Electricity cost brake reduced phase: subsidy capped at 15 ct/kWh for 2,900 kWh; Electricity cost brake: 10 ct/kWh base price, subsidy capped at 30 ct/kWh for 2,900 kWh; Electricity excise reduced from 1.5 to 0.1 ct/kWh; Natural-gas excise reduced from 6.6 to 1.196 ct/m ³	CP0451, CP0452	2022-05-01–2024-12-01	4 of 4 measures use aggregate-equivalent inputs.
Belgium	Social energy tariff and basic electricity package; Social energy tariff and basic gas package; Social tariff for protected residential district-heating customers; Temporary VAT cut on residential electricity from 21% to 6%; Temporary VAT cut on residential natural gas and district heat from 21% to 6%; Temporary excise-duty reduction on petrol and diesel	CP0451, CP0722	CP0452, 2022-03-01–2023-12-01	4 of 6 measures use aggregate-equivalent inputs.
Croatia	District-heating price freeze; Electricity-price mitigation package; Temporary household gas VAT cut from 25% to 5%; VAT cut from 13% to 5% on eligible fresh food products	CP0112, CP0114, CP0116, CP0451, CP0455	CP0113, 2022-04-01–2023-12-01 CP0115, CP0117, CP0452,	2 of 9 measures use aggregate-equivalent inputs.
Cyprus	Additional electricity VAT cut from 9% to 5% for vulnerable tariff-08 consumers; Electricity bill subsidy after expiry of the VAT reduction; Temporary EUR 0.0537/litre excise reduction on heating fuels; Temporary EUR 0.07/litre excise reduction on petrol and road diesel; Temporary electricity VAT cut from 19% to 9% for non-vulnerable consumers	CP0451, CP0722	CP0453, 2021-11-01–2023-02-01	4 of 5 measures use aggregate-equivalent inputs.

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Country	Included price measures	COICOP nents	compo- Assumption window	Input status
Estonia	100% reimbursement of household gas network fees; 50% reimbursement of electricity network fees; 65% compensation of district-heating increase above October 2021 tariff; Deferral of the diesel and liquid-fuel excise increase; Deferral of the electricity excise increase from EUR 1.00 to EUR 1.45/MWh; Household electricity price compensation; Household natural-gas price compensation; Regulated universal electricity service for households - residual layer	CP0451, CP0455, CP0722	CP0452, 2021-10-01–2024-03-01	8 of 8 measures use aggregate-equivalent inputs.
Finland	Temporary 7.5 percentage-point reduction in the biofuel distribution obligation; Temporary electricity VAT reduction from 24% to 10%; Temporary zero VAT rate on domestic passenger transport by air; Temporary zero VAT rate on domestic passenger transport by sea and inland waterway; Temporary zero VAT rate on passenger transport by railway; Temporary zero VAT rate on passenger transport by road and taxi	CP0451, CP0731, CP0733, CP0734	CP0722, 2022-05-01–2023-12-01 CP0732,	1 of 6 measures use aggregate-equivalent inputs.
France	Gas tariff freeze and subsequent gas price cap; Regulated electricity tariff cap and TICFE/TCCFE reduction; Transport-fuel rebate of 18 then 30 cents per litre	CP0451, CP0722	CP0452, 2021-11-01–2023-12-01	3 of 3 measures use direct monthly reference paths; no recalibration.
Germany	Early abolition of the EEG electricity levy (3.723 ct/kWh to zero); Gas price brake; Household electricity price brake: EUR 0.40/kWh on 80% of reference consumption; Temporary VAT cut on natural gas from 19% to 7%; Temporary energy-tax reduction on petrol and diesel	CP0451, CP0722	CP0452, 2022-06-01–2024-03-01	4 of 5 measures use aggregate-equivalent inputs.
Greece	Direct subsidy on automotive diesel; Electricity consumption subsidy for households; Natural-gas consumption subsidy for households	CP0451, CP0722	CP0452, 2021-09-01–2023-12-01	3 of 3 measures use aggregate-equivalent inputs.
Ireland	Electricity Costs Emergency Benefit Scheme credits; Public Service Obligation levy reduction; Temporary VAT cut on electricity from 13.5% to 9%; Temporary VAT cut on gas from 13.5% to 9%; Temporary excise-duty cuts on petrol and diesel	CP0451, CP0722	CP0452, 2022-03-01–2023-09-01	3 of 5 measures use aggregate-equivalent inputs.
Italy	Electricity general system charges removed; Gas general system charges removed or reduced; Strengthened social bonus on electricity bills; Strengthened social bonus on natural-gas bills; Temporary 5% VAT rate on household natural gas; Temporary excise-duty cuts on petrol and diesel	CP0451, CP0722	CP0452, 2021-10-01–2023-12-01	4 of 6 measures use aggregate-equivalent inputs; one measure uses a monthly reference path.

Continued on next page

Country	Included price measures	COICOP nents	compo-	Assumption window	Input status
Latvia	Compensation of 50% of district-heating tariff above EUR 68/MWh; Household compensation for diesel price increases; Household compensation for propane-butane gas price increases; Natural-gas heating compensation of EUR 30/MWh; Removal of OIK and electricity network charges	CP0451, CP0454, CP0722	CP0452, CP0455,	2022-01-01–2023-04-01	5 of 5 measures use aggregate-equivalent inputs.
Lithuania	Compensation of household electricity prices; Partial subsidy on household natural-gas prices	CP0451, CP0452		2022-07-01–2023-06-01	2 of 2 measures use aggregate-equivalent inputs.

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Country	Included price measures	COICOP nents	compo- Assumption window	Input status
Malta	Government subsidy keeping household electricity tariffs stable; Govern- ment subsidy stabilising petrol and diesel retail prices	CP0451, CP0722	2022-01-01–2023-12-01	2 of 2 measures use aggregate- equivalent inputs.
Netherlands	Continuation of reduced petrol, diesel and LPG excise rates, January- June 2023; Electricity price cap EUR 0.40 per kWh up to 2900 kWh; Gas price cap EUR 1.45 per m3 up to 1200 m3; Partial continuation of reduced petrol, diesel and LPG excise rates, July-December 2023; Temporary 21% cut in petrol and diesel excise duties; Temporary VAT cut on energy from 21% to 9% - electricity; Temporary VAT cut on energy from 21% to 9% - gas; Temporary increase in the fixed energy-tax refund - electricity allo- cation; Temporary increase in the fixed energy-tax refund - gas allocation	CP0451, CP0722	CP0452, 2022-01-01–2023-12-01	5 of 9 measures use aggregate- equivalent inputs; 2 measures use monthly reference paths.
Portugal	Bread and cereals zero-VAT basket; Electricity VAT reduction from 13% to 6%; Fish zero-VAT basket; Fruit zero-VAT basket; Household option to return to the regulated natural-gas tariff; ISP and carbon-tax reductions on petrol and diesel; Iberian mechanism limiting gas input costs in elec- tricity generation; Meat zero-VAT basket; Milk cheese and eggs zero-VAT basket; Oils and fats VAT measures; Plant-based drinks VAT measure; Re- duction of electricity network-access tariffs; Vegetables zero-VAT basket	CP0111, CP0113, CP0115, CP0117, CP0451, CP0722	CP0112, 2021-09-01–2026-04-01 CP0114, CP0116, CP0122, CP0452,	11 of 13 measures use aggregate- equivalent inputs; one measure uses a monthly reference path.
Slovakia	Flat household electricity commodity, distribution and system charges; Household heat price cap; Household natural-gas price cap	CP0451, CP0455	CP0452, 2023-01-01–2023-12-01	3 of 3 measures use aggregate- equivalent inputs.
Slovenia	District-heating price cap; Household electricity price cap; Household natural-gas price cap; Regulated seller margin for heating oil; Temporary 50% excise reduction on heating oil; Temporary 50% excise reduction on household electricity; Temporary 50% excise reduction on natural gas for heating; Temporary exemption from electricity network charges; Tempo- rary exemption from natural-gas network charges; Temporary regulated petrol and diesel retail prices	CP0451, CP0453, CP0722	CP0452, 2021-11-01–2023-12-01 CP0455,	10 of 10 measures use aggregate- equivalent inputs.

Continued on next page

Country	Included price measures	COICOP nents	compo- Assumption window	Input status
Spain	Bread and bread flour VAT reduction; Cap on increases in the regulated natural-gas last-resort tariff (TUR); Consolidated electricity social-bonus expansions and enhanced discounts; EUR 0.20 per litre petrol and diesel discount; Electricity VAT reduced from 21% to 10%; Electricity VAT reduced from 21% to 5%; Electricity excise duty reduced from 5.11% to 0.5%; Fresh fruit VAT reduction; Fresh vegetables and pulses VAT reduction; Iberian mechanism limiting gas input costs in electricity generation; Milk cheese and eggs VAT reduction; Natural-gas VAT reduced from 21% to 5%; Olive oil and other edible oils VAT reduction; Temporary regulated gas tariff for homeowners' associations with collective boilers; Temporary suspension of the 7% electricity production value tax (IVPEE)	CP0111, CP0115, CP0117, CP0452, CP0722	CP0114, 2021-06-01–2026-04-01 CP0116, CP0451,	9 of 15 measures use aggregate-equivalent inputs; 2 measures use monthly reference paths.

Notes: Each measure is classified as a tax, subsidy/rebate, or administered-price policy. External monthly paths are used when available; otherwise the reported ratio is the calibrated aggregate-equivalent price factor. Sources: Bruegel Dataset (2021), 'National policies to shield consumers from rising energy prices,' version of 26 June 2023; authors' calculations.

B.1 Austria

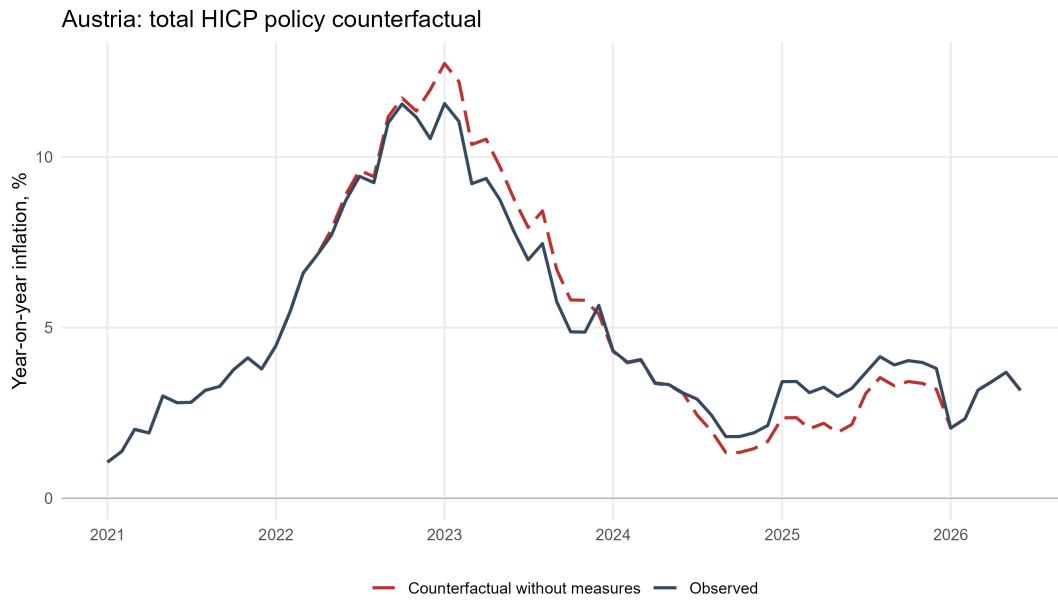


Figure B.1: Austria: observed and counterfactual inflation

B.2 Belgium

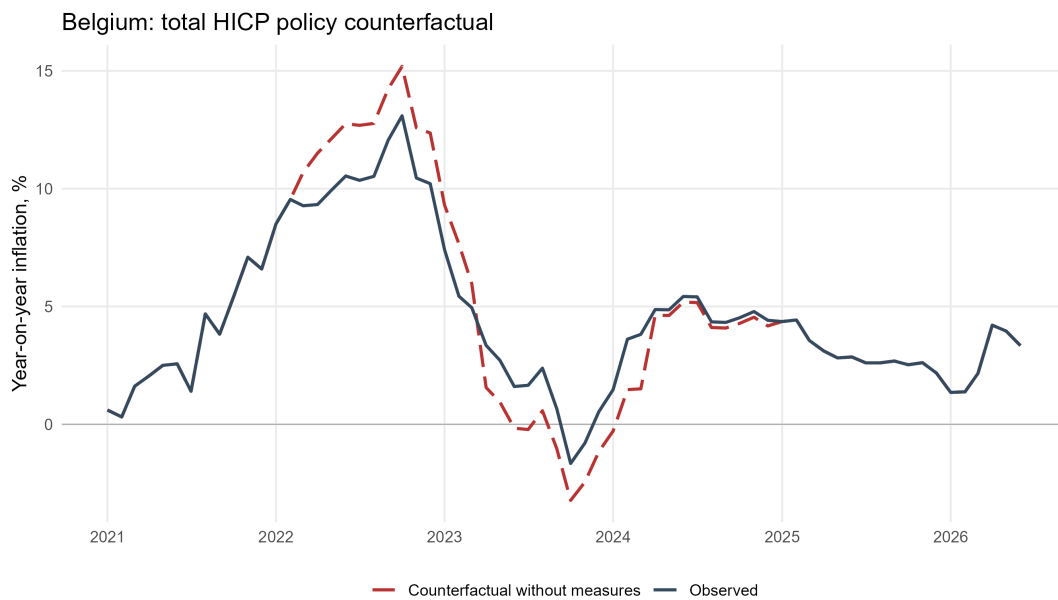


Figure B.2: Belgium: observed and counterfactual inflation

B.3 Cyprus

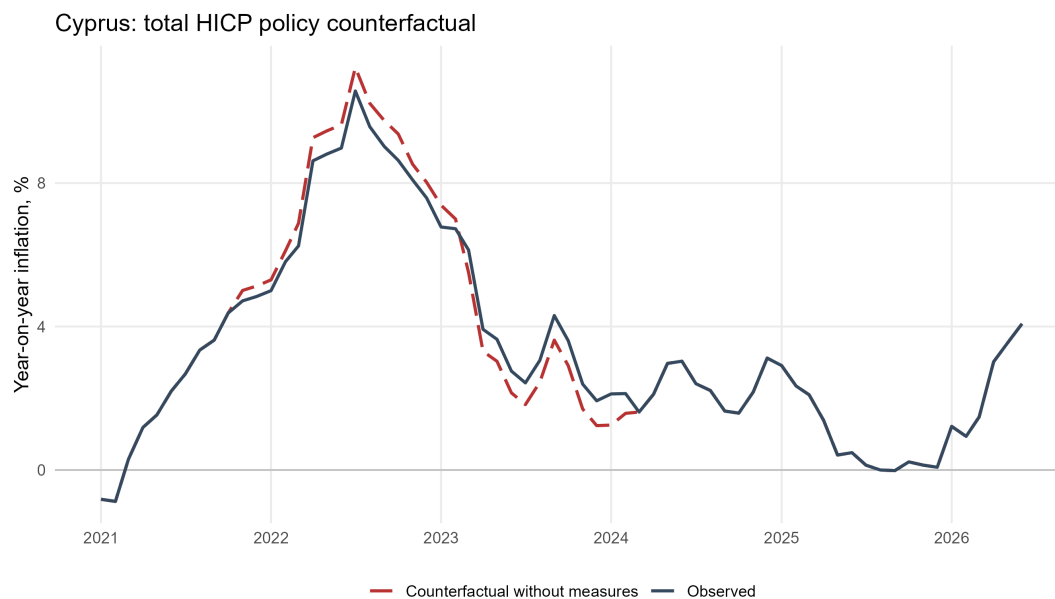


Figure B.3: Cyprus: observed and counterfactual inflation

B.4 Germany

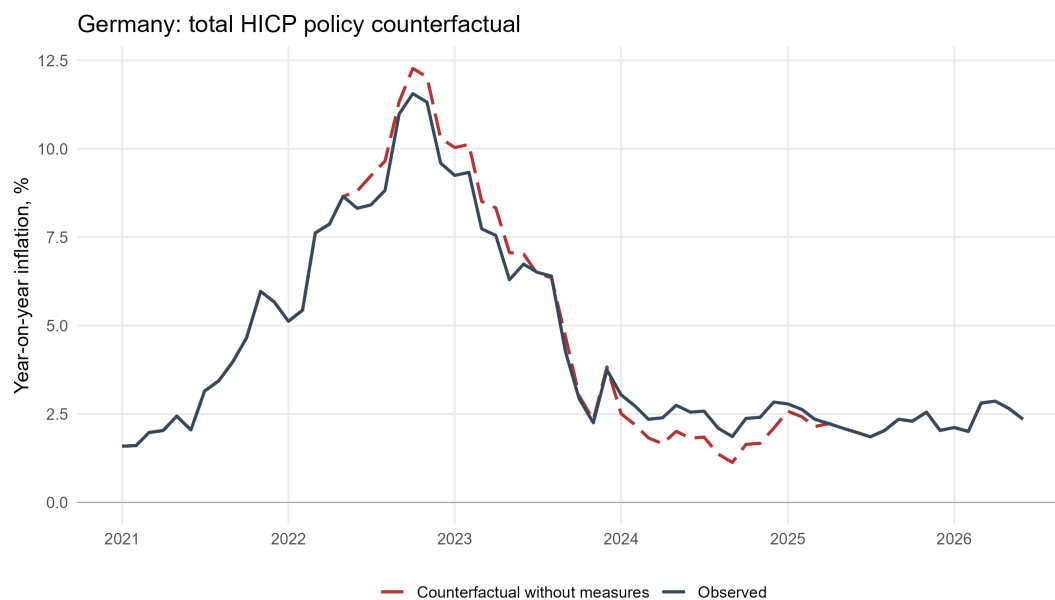


Figure B.4: Germany: observed and counterfactual inflation

B.5 Estonia

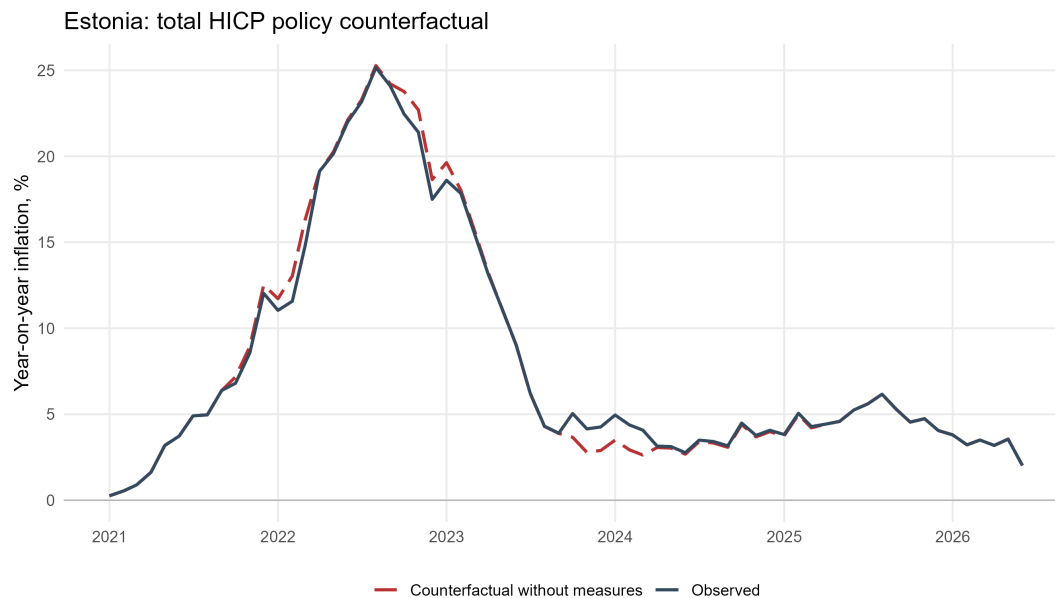


Figure B.5: Estonia: observed and counterfactual inflation

B.6 Greece

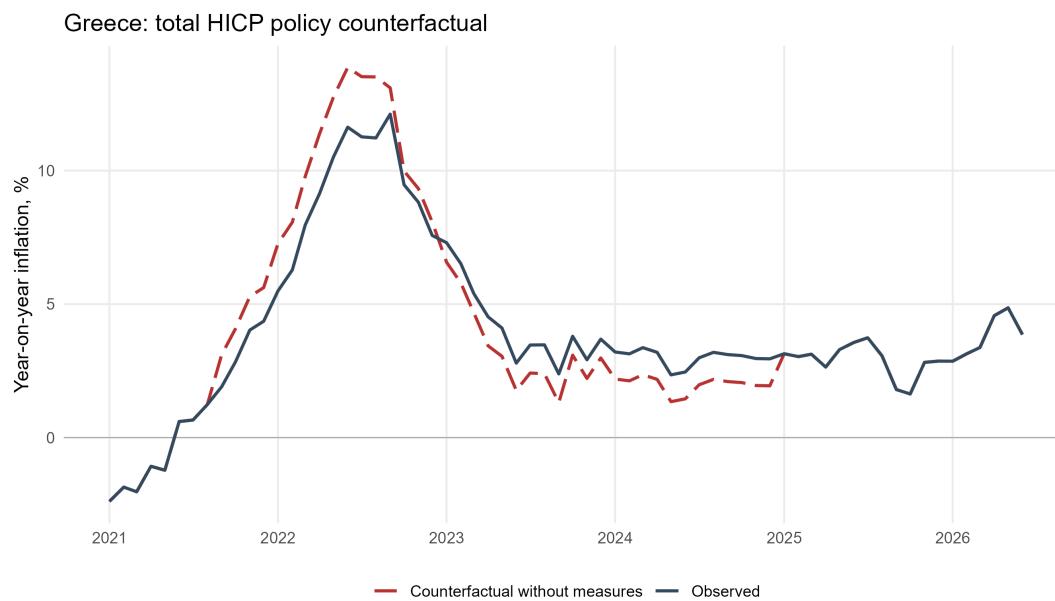


Figure B.6: Greece: observed and counterfactual inflation

B.7 Spain

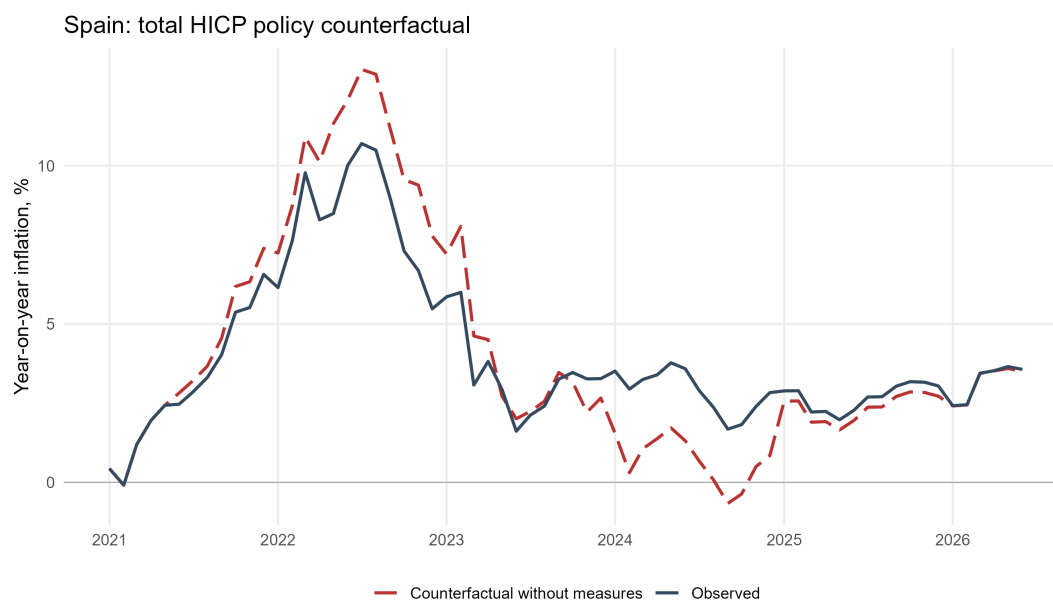


Figure B.7: Spain: observed and counterfactual inflation

B.8 Finland

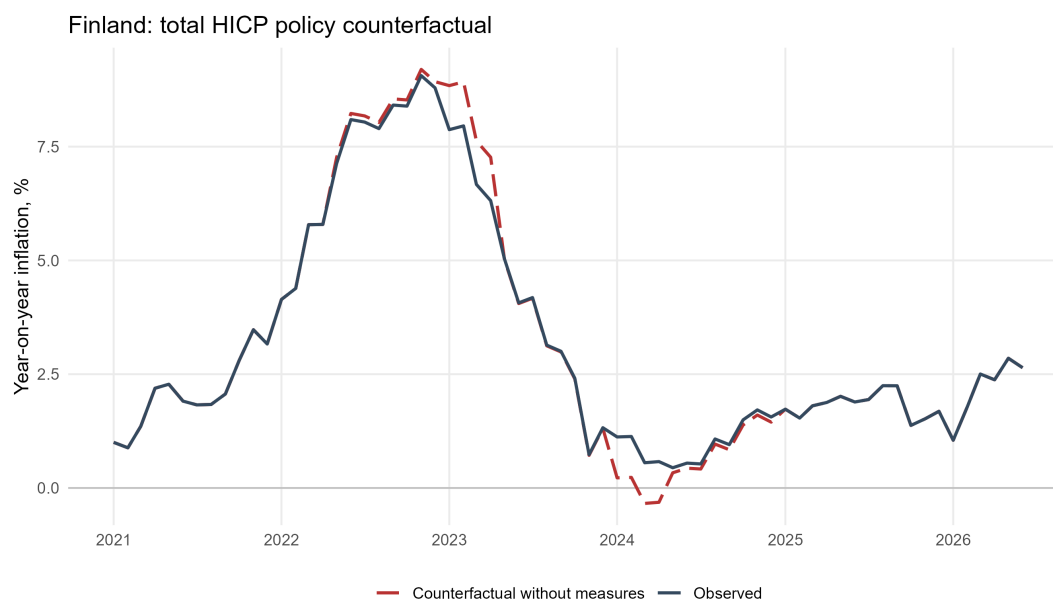


Figure B.8: Finland: observed and counterfactual inflation

B.9 France

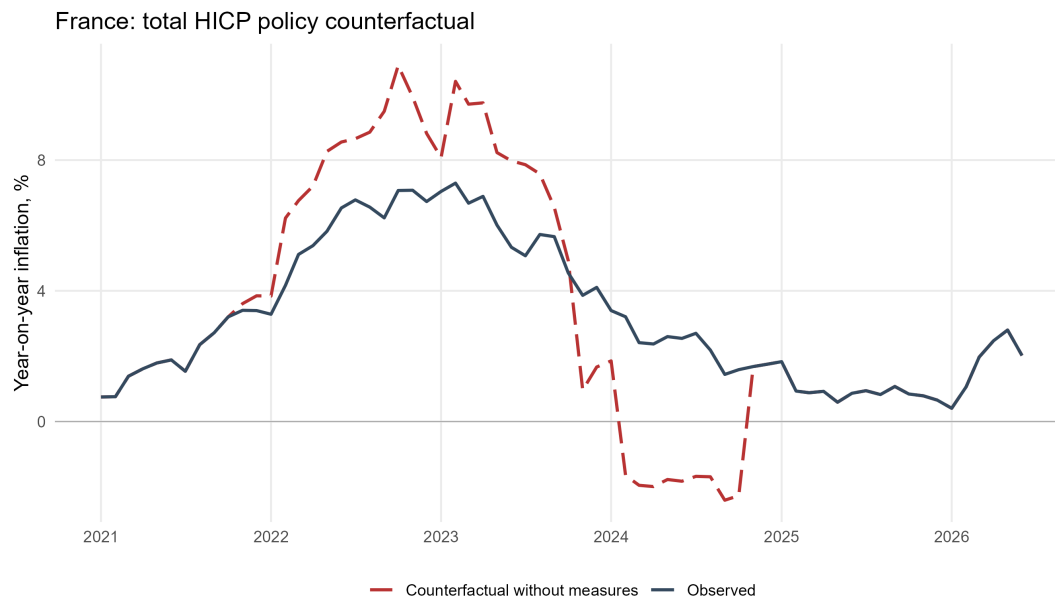


Figure B.9: France: observed and counterfactual inflation

B.10 Croatia

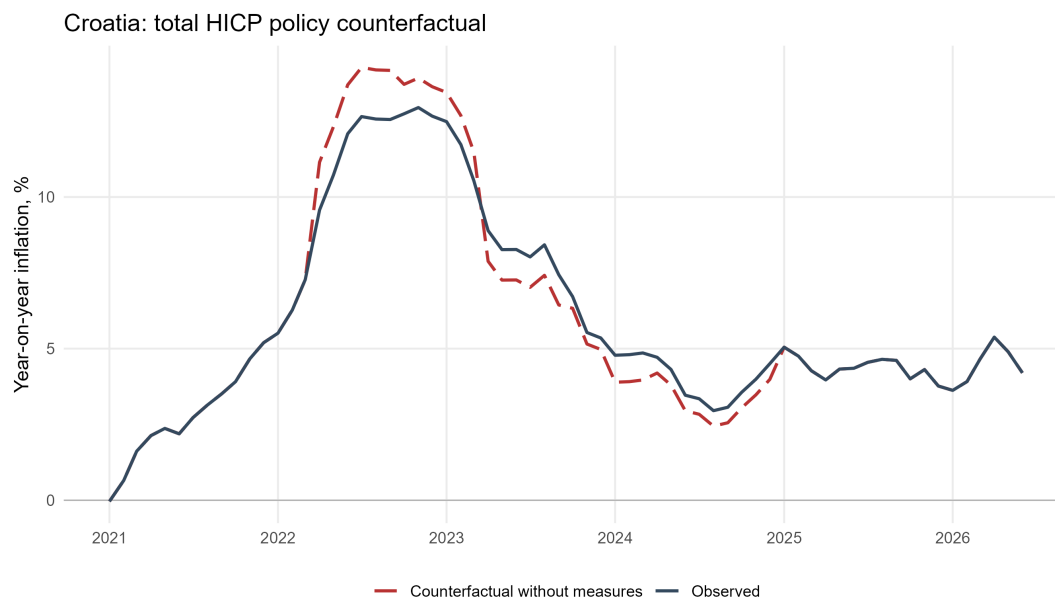


Figure B.10: Croatia: observed and counterfactual inflation

B.11 Ireland

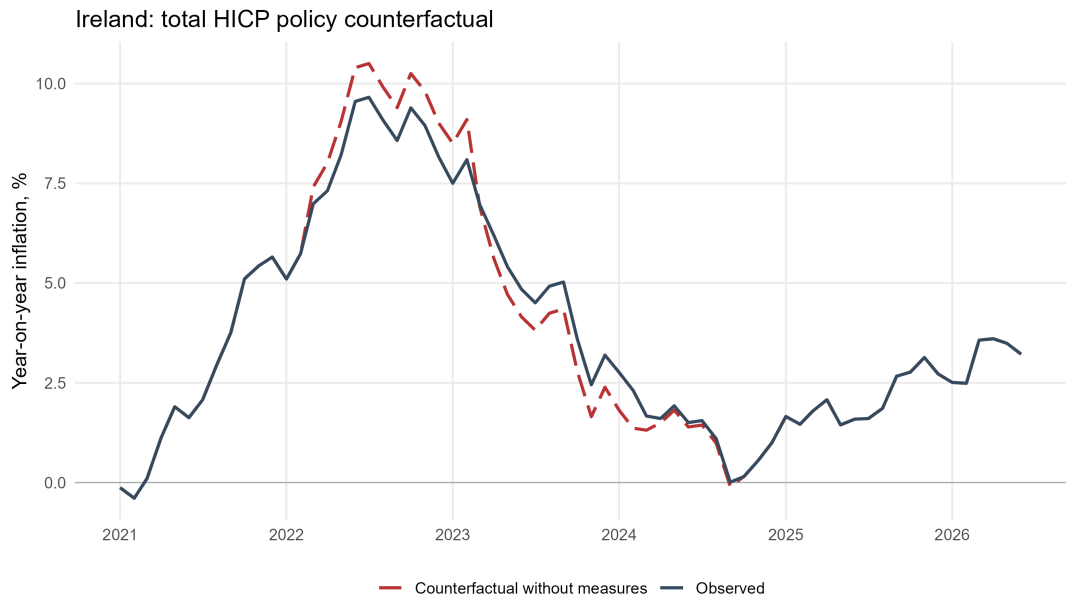


Figure B.11: Ireland: observed and counterfactual inflation

B.12 Italy

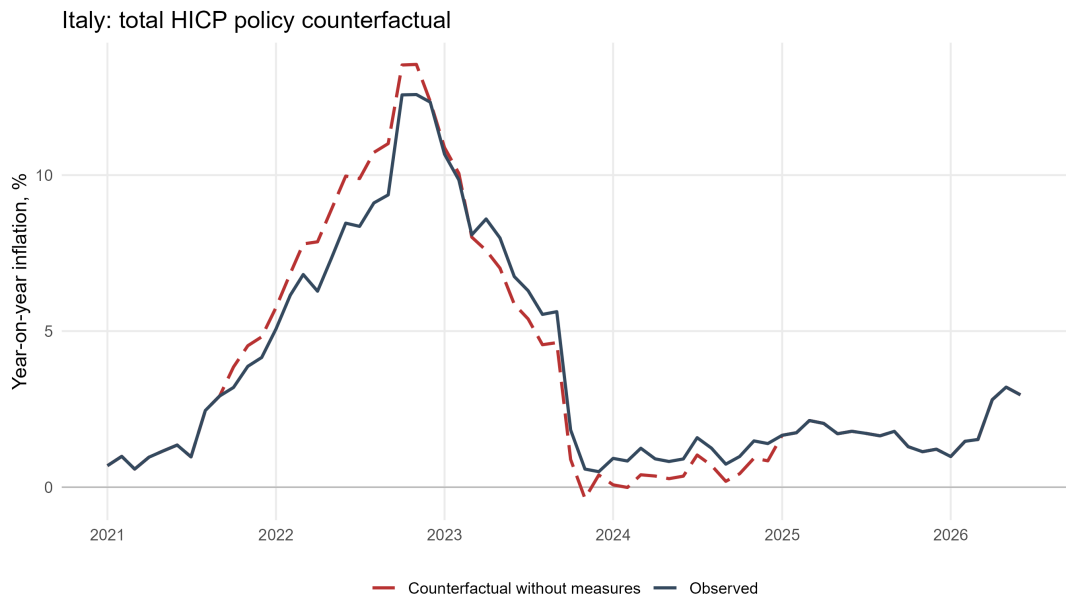


Figure B.12: Italy: observed and counterfactual inflation

B.13 Lithuania

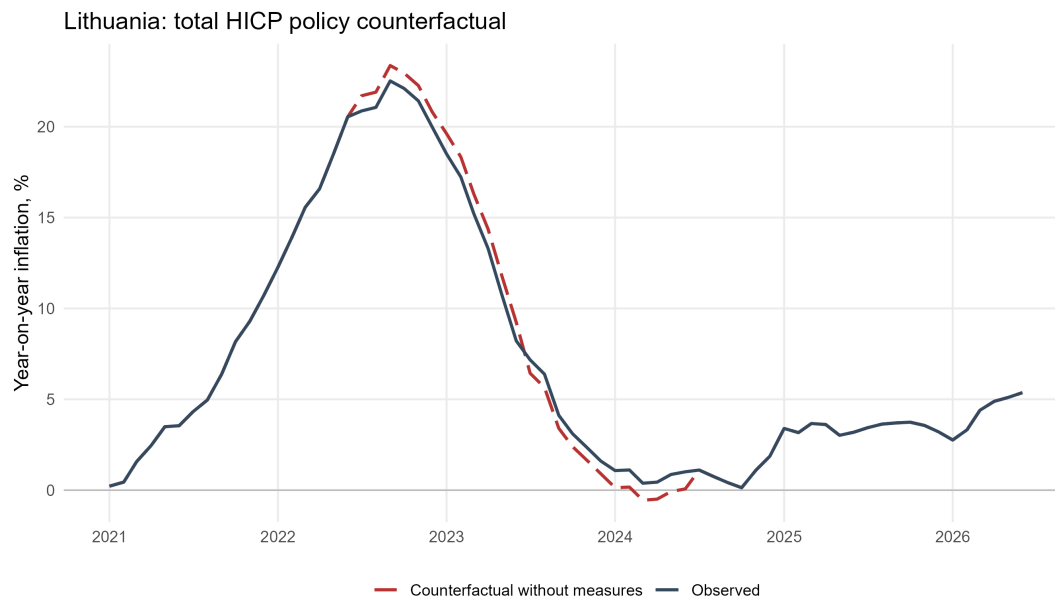


Figure B.13: Lithuania: observed and counterfactual inflation

B.14 Luxembourg

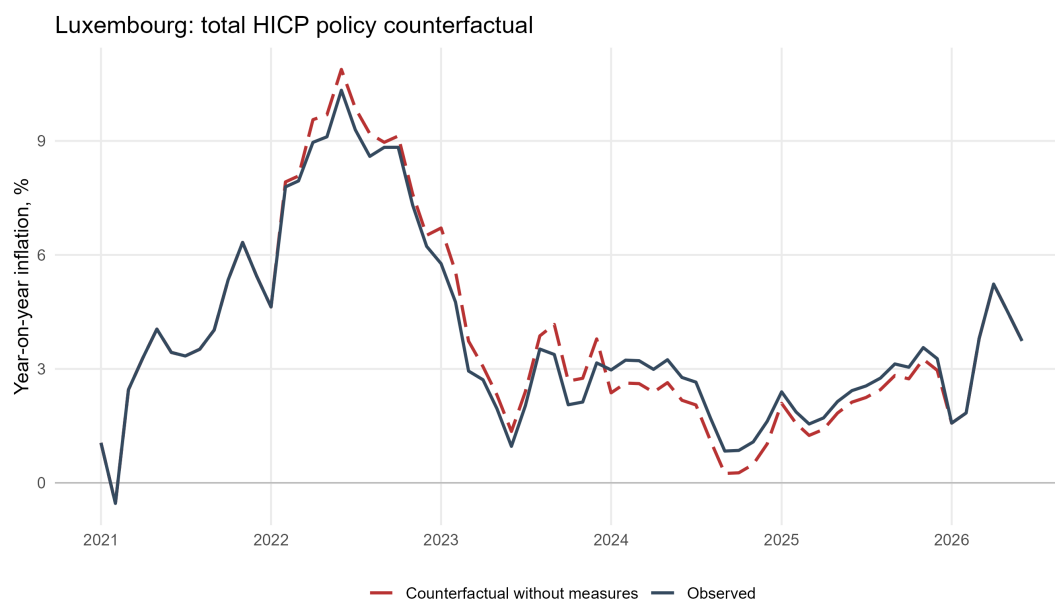


Figure B.14: Luxembourg: observed and counterfactual inflation

B.15 Latvia

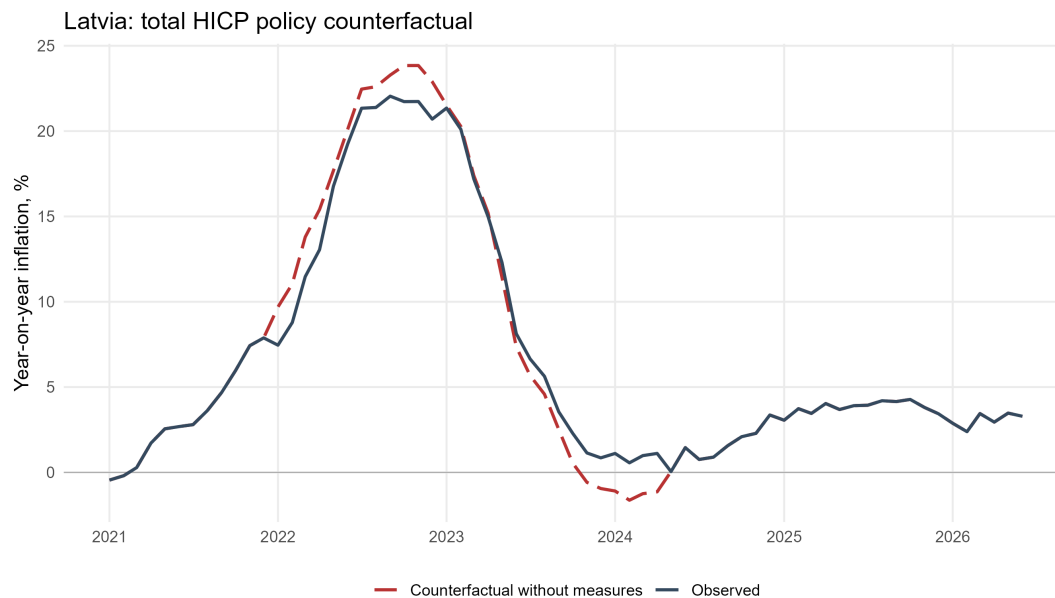


Figure B.15: Latvia: observed and counterfactual inflation

B.16 Malta

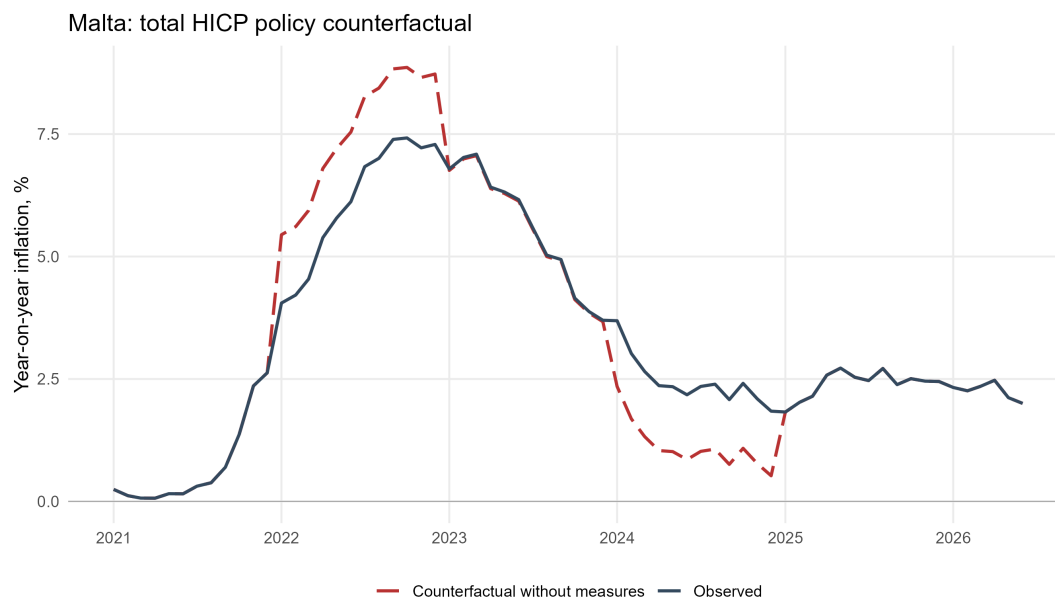


Figure B.16: Malta: observed and counterfactual inflation

B.17 Netherlands

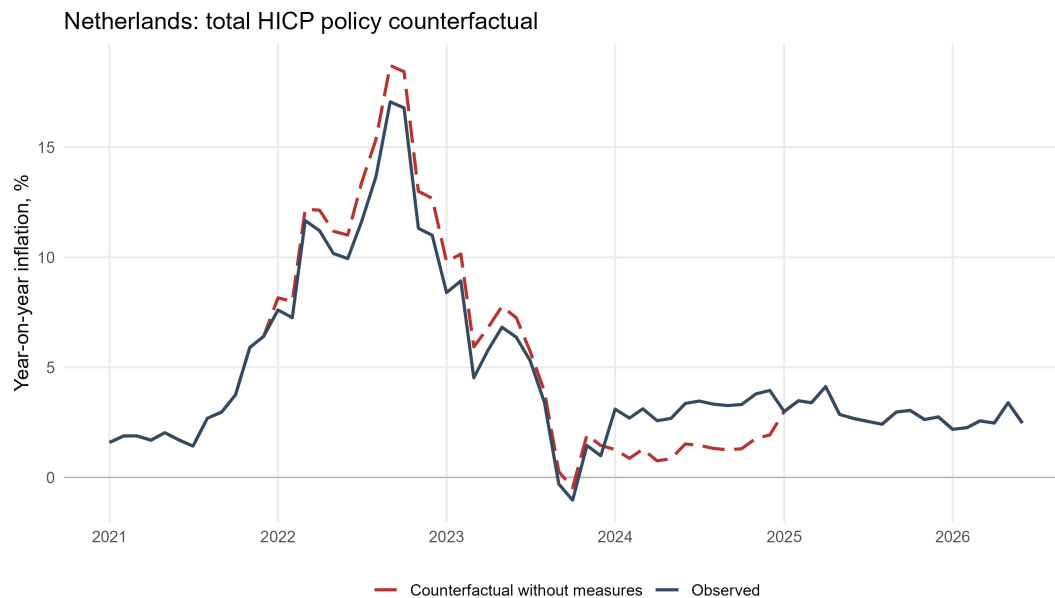


Figure B.17: Netherlands: observed and counterfactual inflation

B.18 Portugal

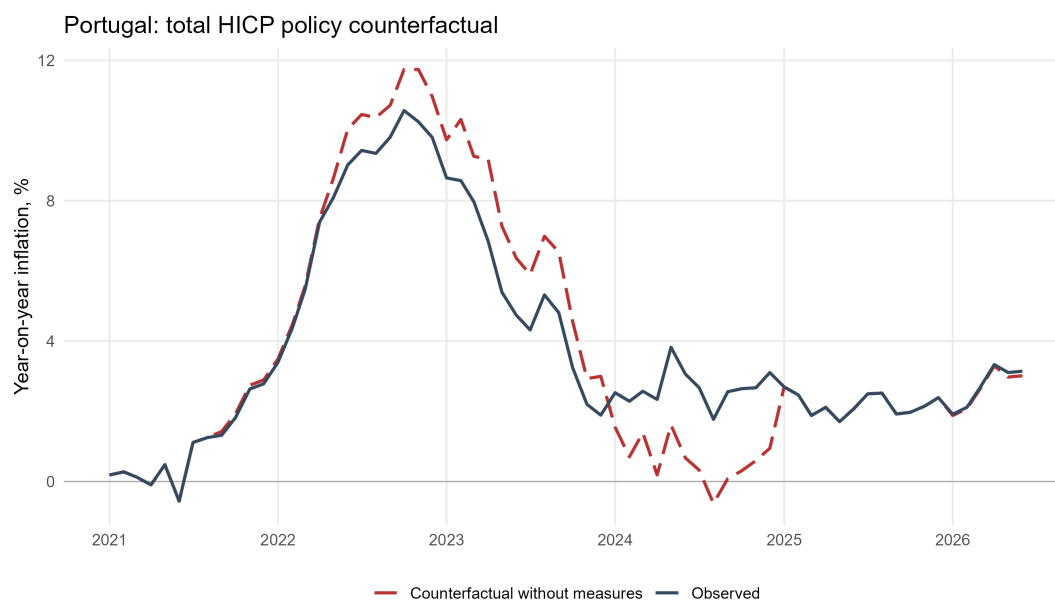


Figure B.18: Portugal: observed and counterfactual inflation

B.19 Slovakia

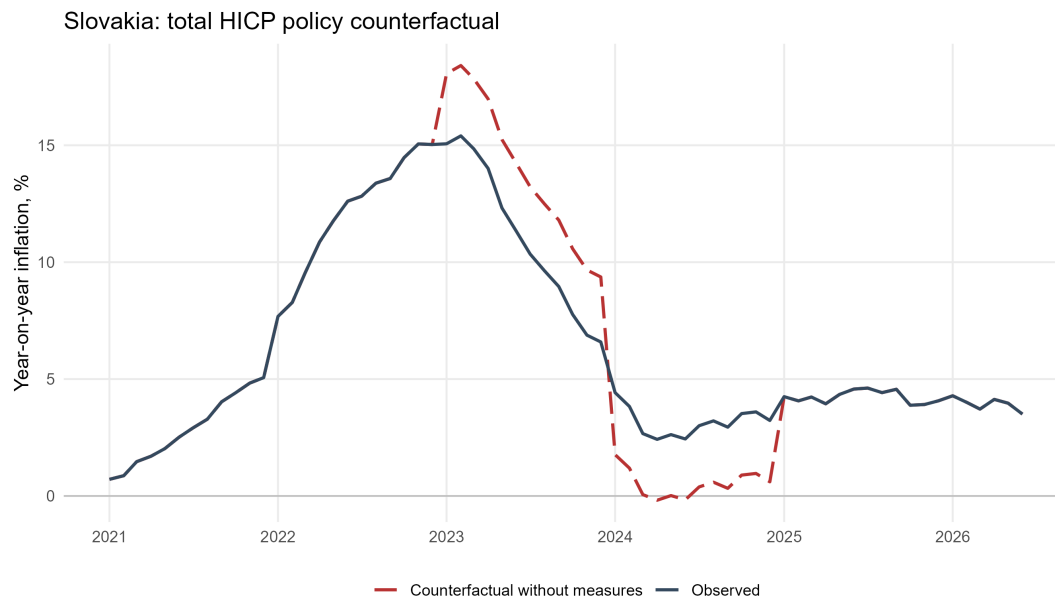


Figure B.19: Slovakia: observed and counterfactual inflation

B.20 Slovenia

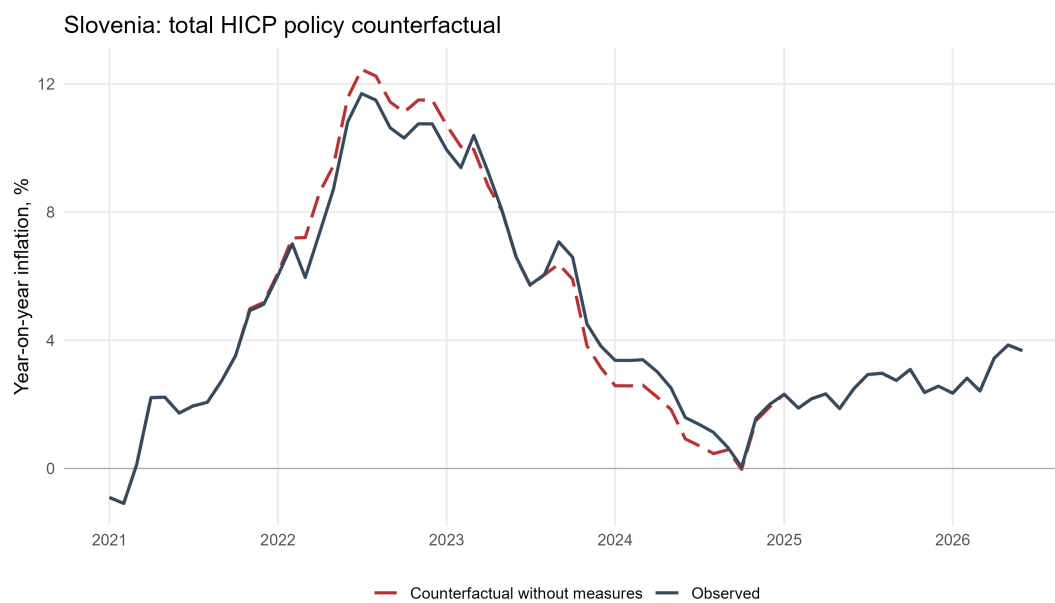


Figure B.20: Slovenia: observed and counterfactual inflation

C The Household Budget Survey in Italy

The most recent complete harmonized Eurostat HBS entry for Italy is 2005. We therefore combine it with the 2015 and 2020 Istat public-use HBS files to construct the household-group baskets.

Age-group baskets are computed directly from the survey-weighted expenditure aggregates in the Istat files, using the age of the household reference person.

For income quintiles, households are ranked by equivalized total consumption using the Carbonaro scale and the survey weights. Because income quintiles are not observed in the target-year files, we transport the relationship between income- and expenditure-quintile baskets observed in 2005. Let E_{jqy} denote expenditure on COICOP group j by expenditure quintile q in year y , and let $s_{jq,2005}^I$ and $s_{jq,2005}^E$ denote the corresponding 2005 shares for income and expenditure quintiles. The estimated income-quintile consumption structure is

$$\hat{s}_{jqy}^I = \frac{E_{jqy} \left(s_{jq,2005}^I / s_{jq,2005}^E \right)}{\sum_k E_{kqy} \left(s_{kq,2005}^I / s_{kq,2005}^E \right)}. \quad (22)$$

Thus, the 2005 income-to-expenditure ratio adjusts each 2015 or 2020 expenditure-quintile component, after which the basket is renormalized to one. A few broad groups are combined to reconcile the two 2005 classifications. The resulting income-quintile structures are estimated rather than directly observed. The procedure addresses the same absence of jointly observed income and consumption discussed by [Donatiello et al. \(2022\)](#).

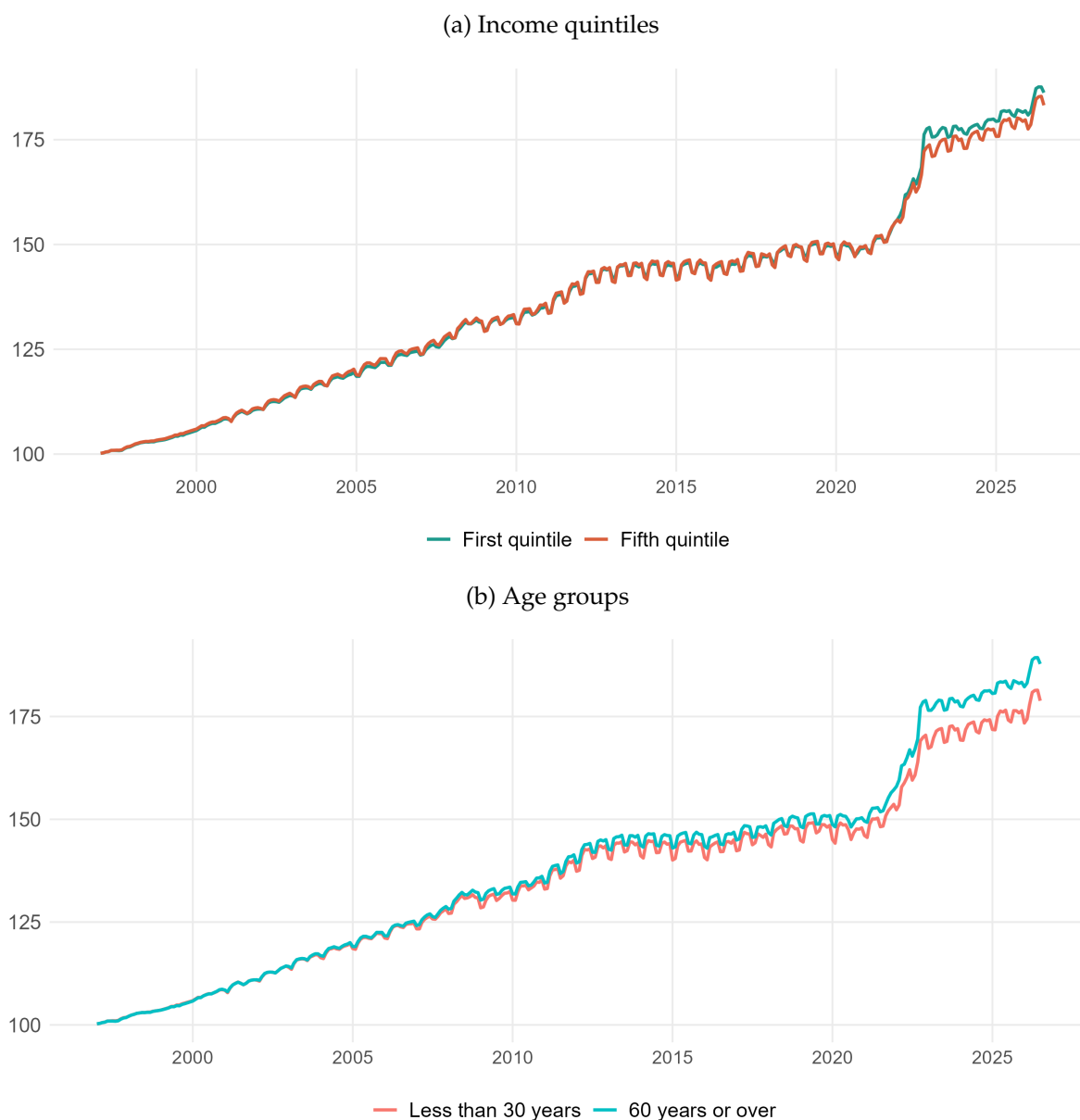
Table [C.1](#) reports the resulting shares of total consumption. Age shares are direct survey estimates; income-quintile shares are bridge estimates. The first and fifth income quintiles account for about 8 and 39 percent of Italian consumption, respectively.

Table C.1: Expenditures by groups (as a share of total expenditures)

Dimension	Group	2015	2020
Age	Less than 30 years	2.0	2.2
Age	From 30 to 44 years	19.7	17.9
Age	From 45 to 59 years	34.8	34.2
Age	60 years or over	43.5	45.7
Income	First quintile	7.8	8.1
Income	Second quintile	12.7	12.8
Income	Third quintile	17.2	17.0
Income	Fourth quintile	23.3	22.8
Income	Fifth quintile	39.0	39.3

Notes. Entries are shares of annual consumption. Age groups follow Eurostat bands; income rows use the Istat–Eurostat quintile bridge. Sources: Istat and Eurostat HBS, authors’ calculations.

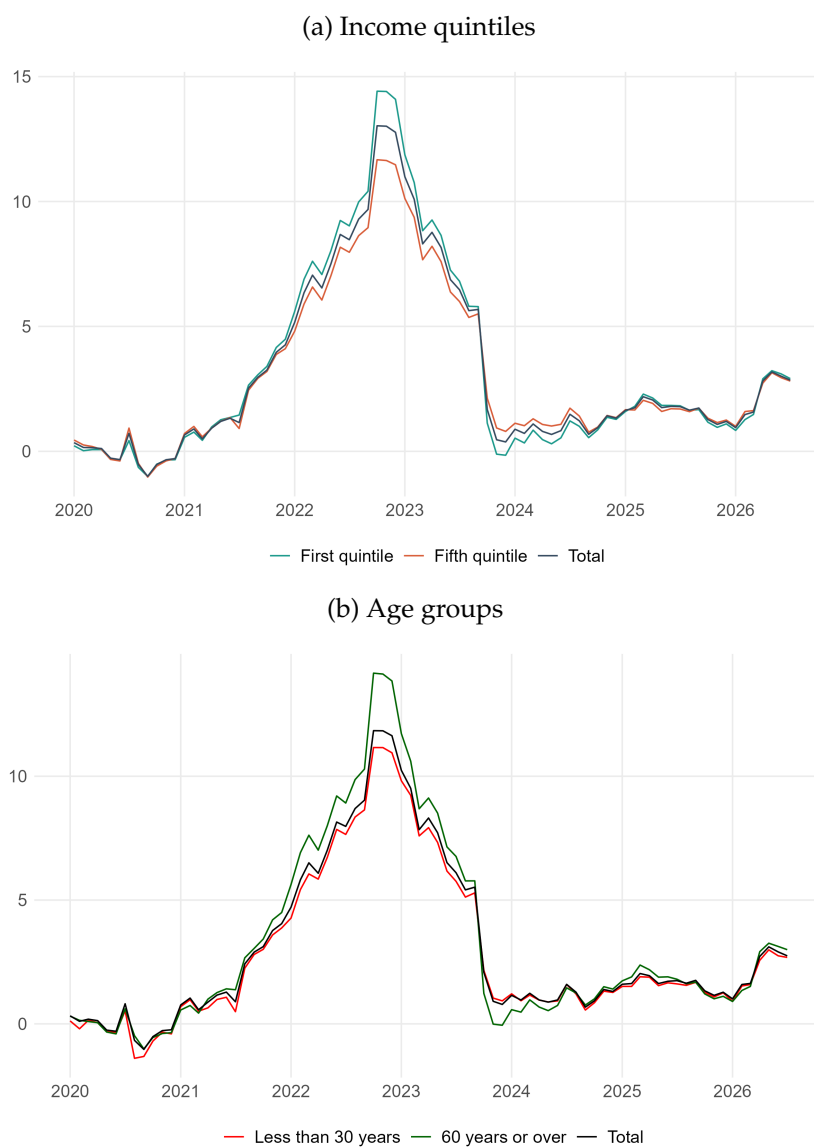
Figure C.3: Price indices by household group, Italy, COICOP digit 3 (index, 1996 = 100)



Notes: The figure reports chained price indices using RAS weighting. Income groups compare the first and fifth quintiles; age groups compare households whose reference person is under 30 with those aged 60 or over.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix C; authors' calculations.

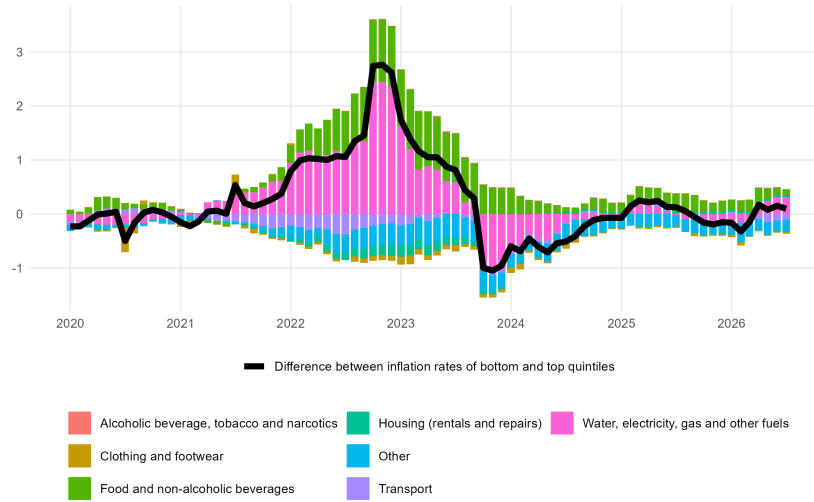
Figure C.4: Inflation by household group, Italy, COICOP digit 3 (in %)



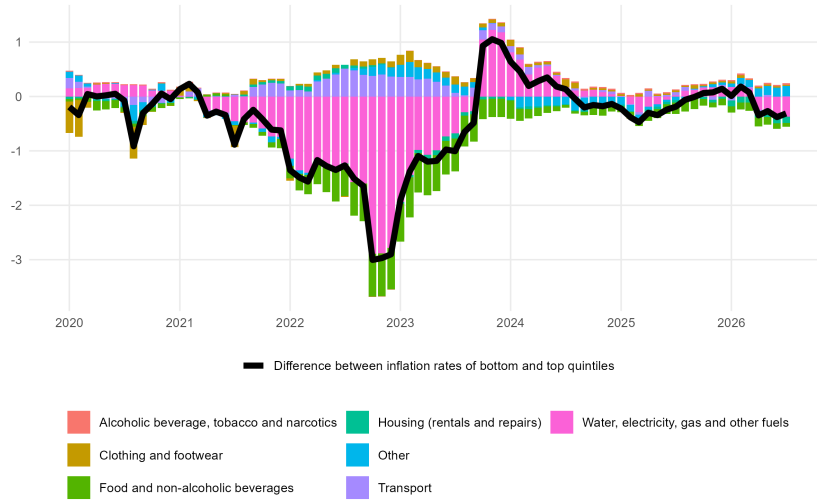
Notes: The figure reports monthly year-on-year inflation rates using RAS weighting for income and age groups.
Sources: Eurostat HICP; Italian HBS inputs described in Appendix C; authors' calculations.

Figure C.5: Contributions to inflation gaps, Italy, COICOP digit 3 (in pp)

(a) Income gap: Quintile 1 minus Quintile 5



(b) Age gap: 60 years or over minus under 30



Notes: Bars decompose the monthly inflation gap into broad consumption components; the black line reports the total gap. Positive values indicate higher inflation for the first group named in each panel.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix C; authors' calculations.

D RAS Weighting

We use the RAS (Row and Column Adjustment Scaling) algorithm. RAS is a bi-proportional matrix scaling method that alternately rescales rows and columns until the required marginal totals are matched ([Bacharach, 1965](#)). In our setting, we combine official HICP item weights with the relative HBS expenditure profile of each household group.

We retain imputed rents in the HBS expenditure profiles even though they are not included in the official HICP. Dropping them before normalizing each group’s expenditures would implicitly treat owner-occupiers as consuming no housing services and would mechanically redistribute the omitted housing share across all remaining items. This is not a neutral rescaling: homeownership, and hence the imputed-rent share, varies systematically across income, age, and residence-area groups. The normalization would therefore inflate the non-housing weights most strongly for groups with high homeownership rates, potentially attributing differences in measured inflation to their consumption patterns when they instead arise from unequal omitted housing shares. [Geerolf \(2023\)](#) shows that this bias can be quantitatively important: because imputed rents are relatively more important for richer, older, and rural households, excluding them can overstate their inflation and can even reverse comparisons between household groups. We therefore preserve imputed rents in the group-level HBS profiles used by the calibration. Their inclusion should be understood as a safeguard against group-specific normalization bias, rather than as a claim that imputed rents belong to the official HICP coverage.

Table [D.1](#) quantifies this sensitivity across countries. The baseline combines HBS actual and imputed rents (041+042) before mapping them to HICP actual rents (041). The first alternative retains only HBS actual rents, while the second removes rents from both the HBS profiles and the HICP basket and renormalizes all remaining items.

Table D.1: Robustness of income-group inflation to the treatment of rents, June 2021–June 2023

Country	HBS 041+042 → HICP 041		HBS 041 → HICP 041		Excluding rents	
	Cumulative inflation	Inflation gap	Cumulative inflation	Inflation gap	Cumulative inflation	Inflation gap
Austria	17.2	0.4	17.1	-0.2	18.3	0.8
Belgium	12.3	-1.1	11.3	-1.6	11.9	-1.1
Croatia	21.4	-0.6	21.1	-0.6	21.3	-0.6
Cyprus	12.0	0.7	12.6	0.6	12.9	0.8
Estonia	33.0	7.5	38.4	7.1	38.8	7.4
Finland	12.5	-1.2	10.0	-3.3	12.8	-0.7
France	12.2	0.2	11.2	-1.3	13.2	0.4
Germany	15.6	1.3	14.7	-1.2	18.4	2.0
Greece	14.7	0.6	14.6	-0.0	15.6	0.8
Ireland	14.9	3.8	17.5	4.4	16.9	4.0
Italy	15.8	2.1	17.2	2.1	17.7	2.3
Latvia	28.9	9.5	36.1	9.6	36.6	9.9
Lithuania	30.4	4.6	33.5	4.5	33.6	4.7
Luxembourg	11.4	-0.2	10.0	-1.5	11.4	-0.2
Malta	12.7	1.6	13.8	1.6	13.0	0.9
Netherlands	16.9	1.2	15.5	-1.7	19.8	2.4
Portugal	14.2	-2.3	12.5	-3.0	13.4	-2.3
Slovakia	25.4	2.7	26.1	2.0	27.3	3.0
Slovenia	18.1	2.4	20.1	2.7	19.7	2.2
Spain	11.8	0.1	11.3	-0.6	12.2	0.2
Euro area	14.7	0.9	14.2	-0.5	16.3	1.3

Notes. Cumulative inflation refers to Q1; the gap is Q1 minus Q5. The baseline combines actual and imputed rent expenditure; alternatives use actual rent only or exclude rent. Entries are percentage points. Sources: Eurostat, authors' calculations.

We distinguish three objects: the HICP product shares, the aggregate product-by-group expenditure mass calibrated by RAS, and the resulting within-group baskets. First, let $s_{g,y}$ denote the share of group g in total consumption expenditure used for HICP weight year y :

$$s_{g,y} = \frac{E_{g,y'}^{\text{HBS}}}{E_{\text{all},y'}^{\text{HBS}}}, \quad \sum_g s_{g,y} = 1, \quad (23)$$

where y' is the HBS wave matched to HICP weight year y . In the implementation, $s_{g,y}$ is selected from the group consumption-share table using the most recent HBS wave available at or before y ; if no prior wave is available, the earliest available wave is used. These targets are expenditure shares, not population shares. For Italy, the income targets use the Istat-based HBS inputs described in Appendix C.

Let $\mathcal{J}_y$ be the set of matched COICOP categories available in the HICP weighting structure in year y . Let $E_{j,g,y'}^{\text{HBS}}$ denote expenditure on item j by group g in the matched HBS wave y' , and let $E_{j,\text{all},y'}^{\text{HBS}}$ denote expenditure on that item by all households. Consistently with the implementation, the HBS correction factor used in the RAS seed is the share of aggregate expenditure on item j accounted for by group g :

$$c_{j,g,y'}^{\text{HBS}} = \frac{E_{j,g,y'}^{\text{HBS}}}{E_{j,\text{all},y'}^{\text{HBS}}}. \quad (24)$$

Thus $c_{j,g,y'}^{\text{HBS}}$ records how expenditure on item j is distributed across household groups. To avoid mixing index-weight units with expenditure shares, define the HICP weight normalized over the matched product set as

$$\bar{\omega}_{j,y}^{\text{HICP}} = \frac{\omega_{j,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_y} \omega_{k,y}^{\text{HICP}}}, \quad \sum_{j \in \mathcal{J}_y} \bar{\omega}_{j,y}^{\text{HICP}} = 1. \quad (25)$$

The initial product-by-group mass is then

$$Z_{j,g,y} = \max \left\{ s_{g,y} \bar{\omega}_{j,y}^{\text{HICP}} c_{j,g,y'}^{\text{HBS}}, \epsilon \right\}, \quad \epsilon = 10^{-12}. \quad (26)$$

The small positive floor is a numerical safeguard to keep the RAS problem strictly non-negative after HICP-HBS matching. RAS transforms this seed into a joint expenditure-share matrix $M_{j,g,y}$. Its group margins equal the groups' expenditure shares, while its product margins equal the normalized HICP shares:

$$\sum_{j \in \mathcal{J}_y} M_{j,g,y} = s_{g,y} \quad \text{for all } g, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}} \quad \text{for all } j. \quad (27)$$

Both sets of margins sum to one, so the constraints are mutually compatible.

The RAS algorithm starts from $M_{j,g,y}^{(0)} = Z_{j,g,y}$ and repeats two scaling steps. At iteration t , the group-margin adjustment is

$$\tilde{M}_{j,g,y}^{(t)} = M_{j,g,y}^{(t-1)} \times \frac{s_{g,y}}{\sum_{\ell \in \mathcal{J}_y} M_{\ell,g,y}^{(t-1)}}, \quad (28)$$

and the product-margin adjustment is

$$M_{j,g,y}^{(t)} = \tilde{M}_{j,g,y}^{(t)} \times \frac{\bar{\omega}_{j,y}^{\text{HICP}}}{\sum_h \tilde{M}_{j,h,y}^{(t)}}. \quad (29)$$

We iterate these two steps by weight year until the maximum absolute row or column marginal error is below 10^{-10} , with a maximum of 1,000 iterations.

The calibrated matrix M is a joint expenditure distribution. The group-specific HICP basket is the corresponding conditional product distribution within group g :

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}} = \frac{M_{j,g,y}}{\sum_{k \in \mathcal{J}_y} M_{k,g,y}}. \quad (30)$$

The equality of the two denominators follows directly from the group-margin constraint. It implies that each group basket sums to one. Moreover, the expenditure-share-weighted average of the group baskets reproduces the normalized HICP product weights:

$$\sum_{j \in \mathcal{J}_y} \omega_{j,g,y} = 1, \quad \sum_g s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}. \quad (31)$$

The package implements the same system on a percentage scale. Internally, the calibrated mass is $M_{j,g,y}^{\text{pct}} = 100M_{j,g,y}$, so its margins are $100s_{g,y}$ and $100\bar{\omega}_{j,y}^{\text{HICP}}$. The package returns these percentage weights in the R variable `weighted_consumption`. Denoting this variable by $w_{j,g,y}^{\text{code}}$, we have

$$w_{j,g,y}^{\text{code}} = \frac{M_{j,g,y}^{\text{pct}}}{s_{g,y}} = 100\omega_{j,g,y}. \quad (32)$$

This quantity therefore sums to 100 within every household group. Thus division by $s_{g,y}$ in the code is the percentage-scale counterpart of dividing $M_{j,g,y}$ by its group-row sum in the normalized mathematical representation.

This preserves the relative information contained in the HBS expenditure profiles while making the aggregate of the household-group baskets consistent with the published HICP weighting structure, up to residual differences induced by product coverage and COICOP matching.

Table D.2 compares the two weights for Germany at COICOP digit 3 over the period beginning in 2019. The RAS calibration tracks the published German HICP more closely than the relative-expenditure weighting, whose implicit aggregate weights are equal across groups.

Table D.2: Germany HICP validation by weighting scheme (differences in pp)

Method	Obs.	Mean diff.	Mean abs. diff.	RMSE	Max abs. diff.
RAS	88	-0.059	0.093	0.118	0.259
Relative expenditure	88	-0.070	0.143	0.166	0.449

Notes: Validation statistics compare the recalculated average of German income-quintile inflation rates with the published all-items HICP inflation rate. The sample is monthly, from January 2019 to April 2026. All differences are recalculated minus published inflation and are reported in percentage points.

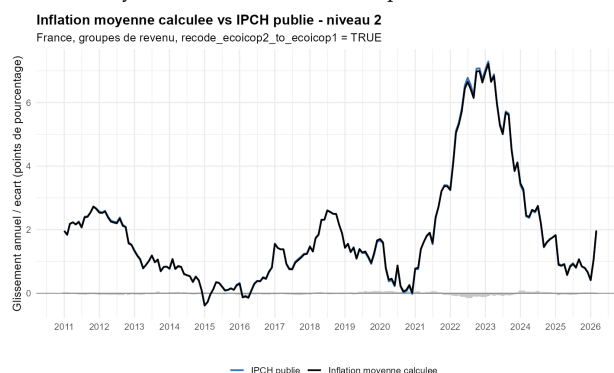
E Validation Tests

This appendix reports three validation checks for the recalculated French inflation series: agreement with published HICP, aggregation of quintile inflation rates, and comparison with official INSEE decile inflation.

E.1 Internal Consistency

Figure E.1 compares COICOP digit-3 RAS recalculated French inflation with published all-items HICP. Figure E.2 compares the total inflation rate with the expenditure-weighted average of income-quintile inflation rates.

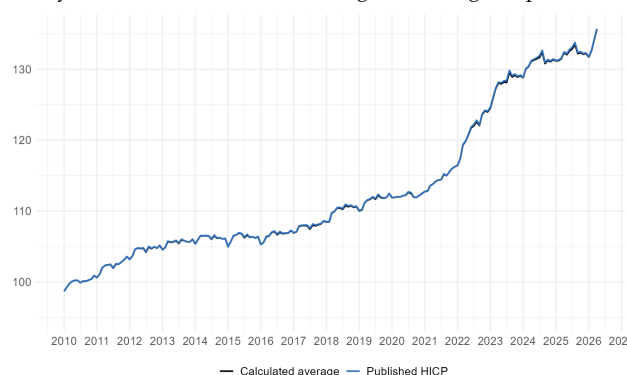
Figure E.1: Internal consistency: France total inflation and published HICP, COICOP digit 3 RAS (in %)



Notes: Blue line: published all-items HICP. Black line: recalculation from 2019 onward. Grey bars: recalculation minus published HICP, in points.

Sources: INSEE, Eurostat, authors' calculations.

Figure E.2: Internal consistency: France total inflation and weighted average of quintiles, COICOP digit 3 RAS (in %)



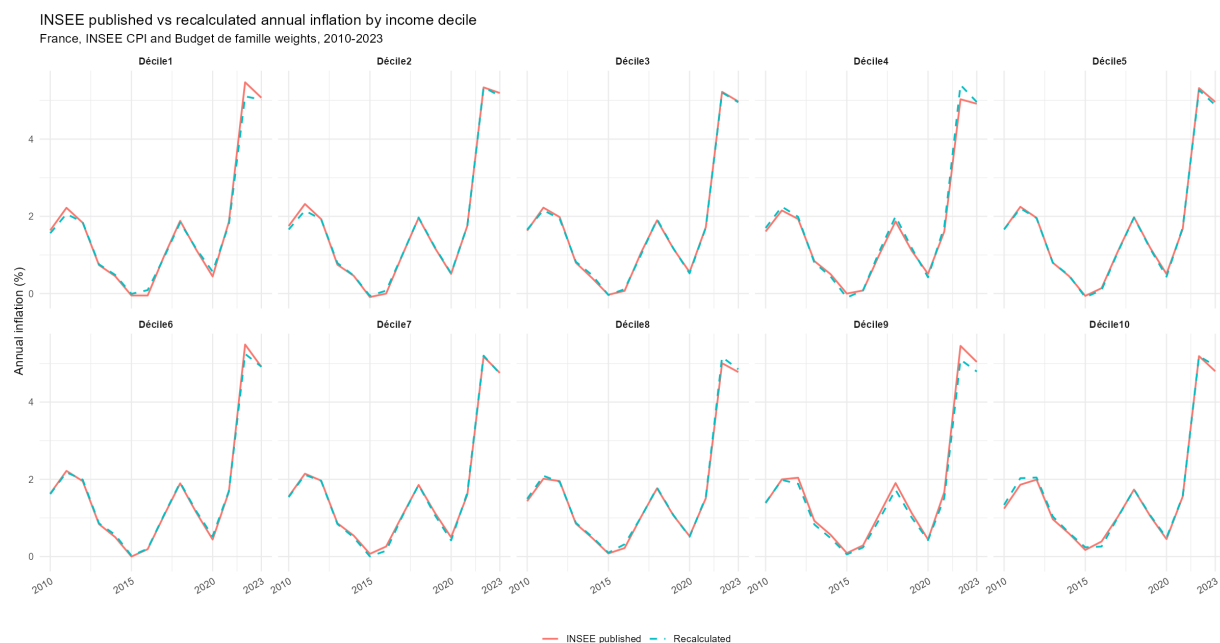
Notes: Recalculated total inflation and the weighted average of recalculated income-quintile inflation rates. Grey bars: weighted average minus total, in points.

Sources: INSEE, Eurostat, authors' calculations.

E.2 External Consistency

Figure E.3 compares the annualized inflation rates computed at COICOP digit 4 with the official INSEE income decile series.

Figure E.3: External consistency: annual inflation by standard-of-living decile, France (in %)



Notes: Official INSEE annual inflation rates by standard-of-living decile are compared with the recalculated series.

Sources: INSEE, Eurostat, authors' calculations.